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### Smart target co-witnessing hit attribution system and method

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#### Abstract

An extended-reality projectile-firing gaming system includes a projectile-firing device that includes a projectile repository, a battlefield object, a network configured to provide communication and control connectivity in accordance with at least one protocol to the projectile-firing device and the battlefield object, an extended-reality gaming application, and an extended-reality device. The extended-reality device includes at least one display and is configured to provide computer-aided vision of real and virtual objects and a computer-aided interface for interaction with the real and virtual objects, the computer-aided interface configured to receive commands or instructions associated with the real and virtual objects and corresponding to the gameplay configuration.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/218,335 filed Jul. 5, 2023, which is a continuation-in-part of U.S. patent application Ser. No. 18/095,911 filed Jan. 11, 2023, now U.S. Pat. No. 11,813,537, which is a continuation-in-part of U.S. patent application Ser. No. 17/862,001 filed Jul. 11, 2022, now U.S. Pat. No. 11,759,717, which claims the benefit of U.S. Provisional Patent Application No. 63/220,343, filed on Jul. 9, 2021. U.S. patent application Ser. No. 18/095,911 also claims the benefit of U.S. Provisional Patent Application No. 63/409,582, filed on Sep. 23, 2022. All applications listed above are incorporated by reference in their entireties herein.

## FIELD OF THE INVENTION

(1) The present invention relates generally to projectile-firing gaming systems and more particularly to a smart target co-witnessing hit attribution system and method.

## BACKGROUND OF THE INVENTION

(2) Indoor or outdoor shooting, blaster or skirmish sports, such as paintball, laser tag, air soft and the like are popular games and hobbies that allow people to run, exercise and have fun. Other shooting exercises, such as hunting, military and police training also involve physical activity. One example of a popular shooting device and game is Gel Blaster. Many people also enjoy video games and, in particular, video games that include guns or shooting (e.g., first-person shooter video games). Unlike video games, however, it is difficult to accurately keep track of hits and how to appropriately attribute a particular hit with a particular shooting device or user. There therefore exists a need to accurately determine when a target or user is hit during a gameplay scenario, particularly when utilizing projectiles. U.S. Pat. Nos. 7,796,155, 9,355,583 and 9,573,064 are incorporated by

reference herein in their entirety.

(3) The background description disclosed anywhere in this patent application includes information that may be useful in understanding the present invention. It is not an admission that any of the information provided herein is prior art or relevant to the presently claimed invention, or that any publication specifically or implicitly referenced is prior art.

#### SUMMARY OF THE PREFERRED EMBODIMENTS

(4) The present invention includes “video gamifying” a real life paintball/gel blaster/combat experience using augmented reality. The present invention in a preferred embodiment includes a system of wirelessly interconnected blasters, targets, wearables, and “battlefield” objects that allow players to engage in precisely tracked and managed projectile sports and games, resulting in an experience of “Video Games Outside”. This system contrasts with projectile-only sports such as airsoft and paintball, which provide tactile feedback but lack automatic score tracking and other digital features. It also contrasts with laser tag or non-projectile-firing sports, which may provide real-time access and control over scoring, game modes, weapon features, etc., but lack tactile feedback, and has to make use of non-intuitive and mostly invisible light physics. In contrast, the present invention includes a system utilizing visible projectile physics and methodology to keep track of score and other gameplay features.

(5) In a preferred embodiment, the present invention includes digitizing an outdoor/indoor blaster gameplay experience thereby providing a mixed reality video game experience through the use of augmented reality (AR), virtual reality (VR), and/or extended-reality (XR) eyewear (goggles, glasses, helmet, etc.) while also including blasters that shoot or fire actual projectiles. It will be appreciated that the projectiles cause the human target a sufficient amount of sensation such that it makes the experience engaging. It will be appreciated that the use of the blaster shown in the accompanying drawings is only exemplary and not limiting. Any type of blaster, gun, etc., that shoots projectiles (e.g., projectile-firing device) is within the scope of the present invention.

(6) The AR/VR/XR eyewear and/or other wearable items, such as vests, pants, belts, shoes, uniforms, jerseys, suits, and the like and the connected system (including all software, programming, architecture, sensors, wireless connectivity, networking, tracking codes, trackers, etc.) allows the participants to integrate video game type elements and graphics (e.g., having a life bar, different types of ammunition, levels, etc.) within a blaster game, experience or training.

(7) In a preferred embodiment, the present invention includes a vest/jersey or other wearable that can sense or determine when a player is hit or struck by a projectile (in an embodiment, at least within certain areas on the outer surface of the wearable). For example, the wearable and connected system can sense or log a hit from the front or the back. In a preferred embodiment, the wearable can log or determine the strength, level or intensity of the impact of the projectile. Alternatively, the application, or a combination of the wearable and the application, can log or determine the strength, level or intensity of the impact of the projectile. Therefore, when a player (or the wearable) is struck from a closer range the player loses a greater amount of health points than from a further range.

(8) In a preferred embodiment, the system can determine which other participant or player shot the player that is hit. This may be done using a location and direction solution in each blaster, e.g., ultra-wide band (UWB) sensors, magnetometer, GPS, WiFi, etc., combined with knowing who is shooting at any given time (so the system can know or determine the approximate time delay between a player firing and when the wearable is hit). The wearable may include buttons or other switches that when pushed or switched cause actions to be taken within the game whether in the AR/VR/XR/virtual environment or in the physical world. For example, pushing a button or touching a predetermined location on the wearable or on a strap or other worn item may cause the player's health points or level to increase after they were previously earned. Any gaming event can be associated with the switch, button or the like.

(9) In a preferred embodiment, a software application (“app” or “application”) is configured to be

usable by all players or participants. The app can provide many different features, including app/game-controlled rate of fire on the blaster. The rate of fire feature (and any other game or invention related feature described herein) can be changeable or upgradeable based on gameplay and other predetermined features. For example, if the player visits a predetermined location and finds a upgraded weapon gun or the like or plays the game for a predetermined length of time, the player may earn a higher rate of fire. If a player has a better or higher level weapon in the game, the player has the ability to cause more damage faster to other players wearing the system connected wearables. The system may include a system-on-chip (SOC) or system-on-module in or on a module connectable to the blaster that connects to the app wirelessly (e.g., via Bluetooth).

(10) The app and system can then control the rate of fire of the blaster. This feature allows for a single blaster to represent many different potential digital weapons in the “game.” For example, a starter weapon might only allow for one shot per second in semi-auto mode and require a four second reload time after ten shots, while the top-end weapon in the game allows for ten shots per second in fully automatic mode and does not impose any “digital reload/time requirements.” As a result, a player would only need to physically reload the blaster every time the physical round hopper is emptied. This will also allow for when a player gets eliminated, their blaster does not fire anymore until a new game starts, they go re-spawn at their base, find an extra life somewhere within the battlefield, or otherwise, or other digital gameplay that is utilized during that game or round. It will be appreciated that there can be multiple scenarios where a player may not be able to use their blaster. For example, in a preferred embodiment of the invention, the system may allow the ability to download different environments (e.g., AR/VR/XR environments) that can be used during gameplay. For example, the environments can include jungles, old west, ships, snow, cities, historic, world monuments or movie, comic book and other related themes (e.g., Star Wars, Marvel, Harry Potter, etc.). These environments may provide reasons or the ability to turn off blasting or provide other features within the system gameplay. For example, in a Harry Potter-related environment, there may be a spell or the like that can be “cast” and prevents all the other teams blasters or weapons from working. In a Star Wars environment, the blaster or other handheld item or weapon may become a lightsaber.

(11) Any type of AR/VR/XR system or eyewear is within the scope of the present invention. For example, the system may connect or combine with Apple's (or others') AR glasses. The use of AR/VR/XR eyewear and the connection to the system, app, etc. may provide for abilities or features for players or participants. For example, the eyewear may provide the ability to see where your teammates are through walls, around buildings, etc. The eyewear may provide the ability to see the health being lost by another participant as you shoot them, and/or see your own health stats at all times or as you are hit. Motion tracking or sensing components, sensors or reflectors on the blaster or other items allow the game to replace the look of the physical blaster or other item in the players hand. In other words, if a player has a pistol in the game, the player (and other players) will see a pistol in the AR/VR/XR eyewear and when the player upgrades to, for example, a bazooka, the gun will graphically change on the screen (e.g., similar to a first person shooter game). For example, the virtual gun shown in AR/VR/XR on the eyewear can change from a pistol to a machine gun when the player upgrades to that gun. The eyewear preferably provides the ability to see and go get supply drops (e.g., weapons, health kit, shield, potions, etc.) in the physical world (e.g., around the yard, playing area, arena, etc.). Accelerometers, gyroscopes or other motion sensing devices can be included on or within the blaster or other physical item to track movement that can be digitized and displayed on the AR/VR/XR eyewear or on the app.

(12) In a preferred embodiment, the battlefield environment provides inflatable or other barricades that are associated with the game (and may be wirelessly connected within the game or gaming session) look like other objects in the game that a player is hiding behind. For example, an inflatable wall may look like a rock wall in AR/VR/XR. QR codes can be used on the barricades and any other object within the game for tracking purposes. The eyewear preferably shows

different “skin” or outfit in the game based on your team, your personal preferences, what skins you own or have earned, etc. The wearables may include QR codes or other motion trackers for improved tracking performance.

(13) In a preferred embodiment, the AR/VR/XR eyewear provides the ability to find virtual shields, first aid kits, better weapons, ammo, treasures, extra lives, etc. at predetermined locations (e.g., laying on the ground, on a barricade or other object). For example, the player sees an object, such as an ammunition magazine in the eyewear at a distance and has to move to the predetermined location to obtain the magazine. Once the system determines the player is at the proper location (e.g., via UWB, magnetometer, GPS, WiFi, etc.), the player obtains the extra ammunition for use in the game. The system preferably provides the ability to create or have a shield around the player so that another player cannot shoot them (the other player's gun may not be able to fire). This can be done via location services and magnetometer data to know someone is pointing at a shielded area and then prevent their blaster from firing.

(14) In an embodiment, a mobile device, such as an iPhone, can be mounted on the side or other location on the physical blaster to allow certain of the system features discussed herein. This can be a version of the system that does not include AR/VR/XR eyewear. Many features described herein can be done without the eyewear. For example, the system connection between the app (via a mobile device on and/or connected to the blaster) can provide the system with features such as app controlled firing, earning of experience and health points, other health kit data, sensing or determination of proximity when firing (i.e., how hard the player is hit by the projectiles), etc. Any or all of the features described herein can be used together to provide a software application experience without the use of AR/VR/XR eyewear.

(15) The system may include an accompanying PC/Mobile/Console game. For example, the applicant may partner with a game like Fortnite or one or more video games can be created or developed that accompany or can be used with the system. Any of the items described herein can be sold or provided as a kit or unit. For example, a gameplay kit can include a number of blasters, inflatable bases and/or barricades, grenades, ammunition, magazines, AR/VR/XR eyewear, downloadable app and/or game that are all sold as a unit. Modules that include components that allow connectivity to or within the system can also be sold or provided. The modules can connect or otherwise attach to items such as blasters. This allows the blaster to be used without being connected to the system (without the module) or within the system. For example, a module can be added to a blaster and a mobile device with software (e.g., a phone with an app) can be secured to the module such that the blaster can be used within the system. In another embodiment, the blaster can include the software and/or app and the necessary connectivity, etc. built therein.

(16) Players may earn experience points (XP) in the system and game by playing outside and entering physical world tournaments. For example, the XP can be earned based on length of playing time or based on points earned on better or worse game play.

(17) An embodiment can include remotely controlled blasters that are included at each teams' base that can be controlled by someone in a different location via the game/system/app (camera and computer controlled firing versus the opposing team).

(18) An embodiment can include players remotely piloting drones that shoot projectiles (e.g., Gel Blaster cannons) or other shooting device and that can attack the other team remotely utilizing a computer or mobile device.

(19) In a preferred embodiment, the system can include features, components or elements for collecting and/or monitoring health data. For example, the system can connect or otherwise be associated with Apple HealthKit, Google Health or other systems to track calories burned while playing, monitor heart rate or determine other health data. In a preferred embodiment, the calories burned and other health data can be used for gamification. For example, the system can include daily, weekly and/or monthly contests for longest/highest game play and calories burned to earn XP. The digitization of the game play experience allows for community tournaments, neighborhood

tournaments, teams, monthly contests, leaderboards, etc. In a preferred embodiment, the system includes the ability to broadcast any of the gameplay via cameras mounted on the players or their equipment (e.g., the eyewear) or by connecting to the feed that is broadcast on any or all of the player's AR/VR/XR eyewear or the app. This allows games to be broadcast or streamed on television, computers, Twitch, Youtube, etc., thereby allowing for a sport, television show, channel, etc. to be based around gameplay, tournaments, etc.

(20) The system can also include remotely operated explosives (e.g., a C4 looking Gel Ball bomb) and inflatable smart forts for each team. The forts or other location/structure may include the ability for the other team to “defeat” or finish the fort. For example, the fort can include sensors or the like so that when the other team shoots a predetermined location or area on the fort (e.g., a “death star hole”) enough times it will cause the inflatable fort/base to deflate.

(21) The system may include built in speed loading of gellets or other projectiles. The system can include the ability to digitally dispense or load ammunition. A smart ammo dispenser may be based on predetermined parameters within the gameplay. For example, a game may be set where players can only refill a certain number of times during the length of the game and after they have reached the limit the system prevents the player from reloading (e.g., locking the canister, stopping the blaster from working, etc.). The right to refill may be earned and the blaster may begin working again.

(22) In a preferred embodiment, the module or other digitally related components of the present invention may be connectable to the power of the blaster. For example, the blaster may include leads or a jack that provides power to the external item (such as a module). This allows a module or other system related component to be retrofitted on a blaster or other similar device that includes a power source. This can provide power so that the direction sensing capability, localization, magnetometer, wideband, IR reader and/or reflector and other components can be powered. The wearable can also include infrared (IR) reflectivity that works or is operatively connected to an IR reader on gun so that the system can sense what player is shooting another player and to provide other capabilities. The system can also provide connection so that other powered accessories (e.g., projectile velocity changer) on the blaster or the barrel thereof can be operated. For example, if a player earns a higher rate of fire or a higher velocity during gameplay, the system can provide instructions (via connectivity) to move or change something mechanically or physically on the gun (e.g., slide a sleeve) to provide the higher velocity projectile firing.

(23) The system may include projectile firing devices, such as sniper rifles, that only work in certain areas or predetermined locations in the field of play. For example, the player may earn a “large sniper rifle” within the game, but the rifle can only be used from a designated location (e.g., a certain hill). In use, based on the system sensing the location of the player the rifle cannot be fired unless the player is in a predetermined location (e.g., on the hill).

(24) It will be appreciated by those of ordinary skill in the art that the present invention allows or provides for the incorporation of elements of arena survival video games into a mixed-reality AR/VR/XR environment. In a preferred embodiment, the system includes guns/blasters that include the capability of connecting to a mobile device that includes an app thereon. In a preferred embodiment, the system includes the ability for players or users to “level up” based on outside playing time, etc. Leveling up can include actions such as the ability to obtain better equipment (e.g., guns, ammunition, etc.), and other aspects desired by the player. This can also utilize or connect to exercise related apps or devices (e.g., Apple Watch, Fitbit, etc.). In a preferred embodiment, the system includes wearables that have sensors or the like thereon that can determine when it is struck. This may include capacitive indicator(s) on the wearable and communication/connectivity to a processor that provides feedback on AR eyewear/glasses or the equivalent. The vest or other armor can also provide sensing of the proximity of a shooter based on impact of the projectile.

(25) In a preferred embodiment, the blaster includes one or more holes or openings that are auto-

adjusted to modify feet-per-second of gel and communication/connectivity to AR interface for different types and levels of blasters. In a preferred embodiment, the system includes Bluetooth or other wireless connectivity between elements of the system that are not physically connected (i.e., vest, glasses, blaster, etc.). Any of the elements or components of the system described herein can be wirelessly connected (or wired). Any of the physical items, e.g., barricades, magazines, forts, bases, wearables, blasters, eyewear and the like can include connectivity to the system.

(26) In a preferred embodiment, the system includes location or other proximity based features (e.g., so a player can see the location of teammates, for providing higher assessed damage based on how close the shooter is to the target, the type of blaster). In a preferred embodiment, the system includes a heads-up-display in AR glasses with mixed reality to accommodate data from vest/blaster and other connected components while engaging with the real, physical environment simultaneously. It will be appreciated that the system can also be utilized for applications in law enforcement, military, training, hunting and other shooting or physical scenarios.

(27) The present invention generally includes a smart target co-witnessing hit attribution system and method. In a preferred embodiment, the present invention includes providing accurate counting of projectile hits on a smart target integrated with an application to keep track of successful hits in order to provide a more rich, immersive, and accurate gaming or shooting experience.

(28) In an embodiment, the smart target system is a stationary, modular device that uses piezoelectric sensors to detect and positively identify impact strikes from gellets (or other projectiles) while filtering out other types of impacts. The smart target includes wireless networking technology to communicate over a mesh network using a particular communications protocol to effectively keep “real-time” or near-real time track of successful hits on the smart target.

(29) In an embodiment, the smart target is a gameplay element in a projectile-firing gaming system that may be affixed, embedded, or modular with wearables, shields, drones, turrets, or other physical objects. Anything that is a physical object within a gameplay environment that a projectile is intended to impact may include a smart target. The smart target preferably includes a piezoelectric (PE) sensor configured to determine an impact of a particular projectile upon (or near) the smart target. One of ordinary skill in the art would understand that a PE sensor is an electromechanical device that uses the piezoelectric effect to measure changes in pressure, strain, force (or other physiological effects) by converting them to an electrical charge. In smart target parlance, the PE sensor is utilized for hit detection.

(30) In accordance with another aspect of the present invention there is provided a smart target that includes a housing for receiving at least two sensors configured to co-witness an impact of an object at or near a predetermined distance from a predetermined location, a first sensor of the at least two sensors configured to detect an impact of an object at or near a predetermined distance from a predetermined location and generate a first sensor signal corresponding to the detected impact, a second sensor of the at least two sensors configured to receive information relating to the detected impact and generate a second sensor signal corresponding to the detected impact, and control circuitry configured to receive the first sensor signal and the second sensor signal, confirm whether the first sensor and the second sensor signals have co-witnessed the same impact of the object by determining whether they were received within a predetermined period of time, and provide a target state update via a network in accordance with at least one protocol.

(31) The control circuitry may be configured to confirm, from the first sensor signal, whether the object includes a projectile based on a trained model configured to recognize impacts of projectiles, and generate projectile impact data when the object is confirmed to be a projectile, wherein the target state update includes the projectile impact data.

(32) The second sensor signal may include one of an infrared signal from an infrared sensor, a brain-computer interface signal from an electroencephalogram sensor, a computer-vision signal from a computer-vision system, a radio-frequency signal from a radio frequency sensor, or a fabric-

deformation signal from a smart textile sensor.

(33) In accordance with another aspect of the present invention there is provided a smart target that includes a first target location, a first piezoelectric sensor configured to detect an impact of an object on or within a predetermined distance from the first target location and generate a first piezoelectric sensor signal corresponding to the detected impact, a first infrared sensor configured to detect infrared radiation at a predetermined wavelength and generate a first infrared sensor signal, control circuitry configured to confirm a successful projectile impact on the first target location when the object is determined to be a projectile by comparing the first piezoelectric sensor signal with a piezoelectric impact signal model trained to detect impact of the projectile and when an impact time of the object and a detection time of the infrared radiation each occurred within a first predetermined period of time, generate first target location state data indicating that the first target location was successfully impacted by the projectile, and transmit the first target location state data via a network in accordance with at least one protocol.

(34) The first piezoelectric sensor signal may be filtered before it is compared with the piezoelectric impact signal model. The first infrared sensor signal may include an angle of detection and an intensity of infrared radiation.

(35) The first piezoelectric sensor may be further configured to detect the impact of the object on or within a predetermined distance from a second target location.

(36) The smart target may further include a second target location, a second piezoelectric sensor configured to detect an impact of an object on or within a predetermined distance from the second target location and generate a second piezoelectric sensor signal corresponding to the detected impact, and a second infrared sensor configured to detect infrared radiation at a predetermined wavelength and generate a second infrared sensor signal.

(37) The control circuitry may be further configured to confirm a successful projectile impact on the second target location when the object is determined to be a projectile by comparing the second piezoelectric sensor signal with the piezoelectric impact signal model trained to detect impact of the projectile and when an impact time of the object and a detection time of the infrared radiation each occurred within the first predetermined period of time, generate second target location state data indicating that the second target location was successfully impacted by the projectile, and transmit the second target location state data via a network in accordance with at least one protocol.

(38) The smart target may include a frame configured to receive the first target location. The first target location may be located on a wearable device.

(39) The first target location state data may be configured to be received by an extended-reality gaming application configured to control the projectile-firing device and the wearable device in accordance with a gaming session having a gameplay configuration for providing an extended-reality environment, receive projectile-firing device data from the projectile-firing device, the projectile-firing device data comprising an orientation and location of the projectile-firing device and a time at which the projectile-firing device was fired, update gaming metric data to indicate a successful impact of the fired projectile on the first target location and attribute a successful hit to the projectile-firing device on the first target location when the time the impact occurred on the first target location and the time the projectile-firing device was fired occur within a second predetermined period of time.

(40) The gaming metric data may be updated to indicate the successful impact of the fired projectile on the first target location and the successful hit is attributed to the projectile-firing device on the first target location when the location of the projectile-firing device is within a predetermined range from the first target location and the orientation of the projectile-firing device is aligned at a predetermined area associated with the first target location. The updated gaming metric data may be configured to be displayed to a user.

(41) In accordance with another aspect of the present invention there is provided a smart target co-witnessing hit attribution system that includes a network configured to provide communication and



control connectivity in accordance with at least one protocol, a projectile-firing device comprising a projectile repository configured to carry projectiles for firing and an infrared emitter configured to emit infrared radiation at a predetermined wavelength and beamwidth, a smart target that includes a piezoelectric sensor configured to detect an impact of an object at or near a predetermined distance from a predetermined location relative to the smart target and generate a piezoelectric sensor signal corresponding to the detected impact, an infrared sensor configured to detect infrared radiation at a predetermined wavelength and generate an infrared sensor signal, control circuitry configured to confirm a successful projectile impact on the smart target when the object is determined to be a projectile by comparing the piezoelectric signal with a predetermined piezoelectric impact signal model trained to detect impact of the projectile and the impact of the object and the detection of the infrared radiation occurred within a first predetermined period of time, generate target state data comprising the successful projectile impact, and provide the target state data via the network, an extended-reality gaming application configured to control the projectile-firing device and the wearable device in accordance with a gaming session having a gameplay configuration for providing an extended-reality environment, receive projectile-firing device data from the projectile-firing device, the projectile-firing device data comprising an orientation and location of the projectile-firing device and a time at which the projectile-firing device was fired, receive the target state data from the smart target, the target state data comprising a time the smart target was impacted by the projectile, update gaming metric data to indicate a successful impact of the fired projectile on the smart target and attribute a successful hit to the projectile-firing device on the smart target when the time the impact occurred on the smart target and the time the projectile-firing device was fired occur within a second predetermined period of time.

(42) The gaming metric data may be updated to indicate the successful impact of the fired projectile on the smart target and the successful hit is attributed to the projectile-firing device on the smart target when the location of the projectile-firing device is within a predetermined range from the smart target and the orientation of the projectile-firing device is aligned at a predetermined area associated with the smart target.

(43) The network may be a target mesh network configured to transmit messages including the target state data to the extended-reality gaming application. The at least one protocol may be configured to effectuate transmission and reception of data to nodes within the network using hexadecimal values. The hexadecimal values may specify data concerning a status of the target. The hexadecimal values specify data concerning a status of a user.

(44) In accordance with another aspect of the present invention there is provided a smart target co-witnessing hit attribution method. A first sensor signal is received corresponding to detection of an impact of an object at or near a predetermined distance from a predetermined location relative to a first smart target location, a second sensor signal relating to the detected impact is received; the first sensor signal and the second sensor signal are determine to be received within a first predetermined period of time, and a target state update is provided via a network in accordance with at least one protocol.

(45) From the first sensor signal, it is determined that the object comprises a projectile based on a trained model configured to recognize impacts of projectiles, and projectile impact data is generated, wherein the target state update includes the projectile impact data.

(46) The method may further include wherein the target state update is received, projectile-firing device data from a projectile-firing device is received, the projectile-firing device data comprising an orientation and location of the projectile-firing device and a time at which the projectile-firing device was fired, and gaming metric data is updated to indicate a successful impact of the fired projectile on the first smart target location and attribute a successful hit to the projectile-firing device on the first smart target location when the time the impact occurred on or near the first smart target location and the time the projectile-firing device was fired occur within a second

predetermined period of time.

(47) The gaming metric data may be updated to indicate the successful impact of the fired projectile on the first smart target location and the successful hit is attributed to the projectile-firing device on the first smart target location when the location of the projectile-firing device is within a predetermined range from the smart target and the orientation of the projectile-firing device is aligned at a predetermined area associated with the first smart target location.

(48) In accordance with another aspect of the present invention there is provided an extended-reality projectile-firing gaming system. The system includes a projectile-firing device comprising a projectile repository, a battlefield object, a network configured to provide communication and control connectivity in accordance with at least one protocol to the projectile-firing device and the battlefield object, an extended-reality gaming application, and an extended-reality device. The extended-reality gaming application is configured to control the projectile-firing device and the battlefield object in accordance with a gaming session having a gameplay configuration for providing an extended-reality environment, receive projectile-firing device data from the projectile-firing device, the projectile-firing device data indicating that a projectile was fired, and receive battlefield object data from the battlefield object, the battlefield object data indicating that an impact occurred on the battlefield object. The extended-reality device includes at least one display and is configured to provide computer-aided vision of real and virtual objects and a computer-aided interface for interaction with the real and virtual objects, the computer-aided interface configured to receive commands or instructions associated with the real and virtual objects and corresponding to the gameplay configuration.

(49) The computer-aided interface may include voice-activation control configured to select one or more of the real and virtual objects and to provide the commands or instructions.

(50) The computer-aided interface may include an eye sensor configured to monitor eye contact with the real and virtual objects.

(51) The commands or instructions may be selectable using a control input when the eye contact is aligned with one of the real and virtual objects.

(52) The eye sensor may identify a selection of the commands or instructions associated with the real and virtual objects when the eye contact is fixed on one of the real and virtual objects for at least a first predetermined time period.

(53) The eye sensor may identify a selection of the commands or instructions associated with the real and virtual objects when the eye contact is fixed on one of the real and virtual objects for at least a first predetermined time period but does not exceed a second predetermined time period.

(54) The control input may be voice-activated. The control input may include at least one sensor configured to monitor hand movements and the commands or instructions are selectable based on the monitored hand movements.

(55) The control input may be a remote device. The remote device is coupled to the projectile-firing device. The remote device may be configured to receive data via the network from at least one of the projectile-firing device and the battlefield object. The remote device may be configured to be affixed to the projectile-firing device and provide at least one of near-field communication, radio frequency, infrared, magnetometer, and global positioning satellite sensory data to the extended-reality gaming application.

(56) The real and virtual objects may be displayed in response to the projectile-firing device data. The real and virtual objects may be displayed in response to the battlefield object data.

(57) The gameplay configuration may include information corresponding to at least one of a game mode, a health meter, a shield meter, a projectile tally, a projectile repository tally, a rate of fire, a reload interval, a virtual power up, a success tally, a failure tally, a success to failure ratio, an alert, a battlefield role, a team affiliation, a battlefield objective, and a set of battlefield parameters.

(58) The type of the projectile-firing device may be selectable through the computer-aided interface. The commands and instructions may include one or more of an amount of projectiles

available for firing, a predetermined velocity of the projectiles for firing, a predetermined rate of fire of the projectiles, and a predetermined reload interval of the amount of projectiles.

(59) The player may be selectable through the computer-aided interface and the extended-reality gaming application may facilitate communication with the player. The virtual objects may be sharable through the computer-aided interface with a player. The extended-reality device may be configured to display a prompt to provide at least one of the commands or instructions. The virtual objects may include a menu of selectable inputs corresponding to the commands or instructions. The menu of selectable inputs may be a drop-down menu.

(60) The virtual objects may depict an application configured to be executed on the extended-reality projectile-firing gaming system.

(61) The extended-reality gaming application may be further configured to update gaming metric data to indicate a successful impact of the fired projectile on the battlefield object and a successful hit by the projectile-firing device on the battlefield object when a time the impact occurred on the battlefield object and a time the projectile was fired occur within a predetermined time period, wherein the predetermined time period is selected to confirm that the projectile-firing device fired the fired projectile that was detected as impacting the battlefield object, and wherein the updated gaming metric data is displayed on the extended-reality device and configured to modify the real and virtual objects in real or near real-time.

(62) The updated gaming metric data may be further configured to receive a location and orientation of the projectile-firing device, wherein the extended-reality gaming application records a successful impact of the fired projectile on the battlefield object and a successfully fired projectile by the projectile-firing device associated with the battlefield object when the location of the projectile-firing device is within a predetermined range from the battlefield object and the orientation of the projectile-firing device is aligned at a predetermined area associated with the battlefield object.

(63) The battlefield object may include a smart target comprising a target location, a piezoelectric sensor configured to detect an impact of an object on or within a predetermined distance from the target location and generate a piezoelectric sensor signal corresponding to the detected impact, an infrared sensor configured to detect infrared radiation at a predetermined wavelength and generate an infrared sensor signal, and control circuitry configured to confirm a successful projectile impact on the target location when the object is determined to be a projectile by comparing the piezoelectric sensor signal with a piezoelectric impact signal model trained to detect impact of the projectile and when an impact time of the object and a detection time of the infrared radiation each occurred within a predetermined period of time, generate target location state data indicating that the target location was successfully impacted by the projectile, and transmit the target location state data via the network in accordance with at least one protocol, wherein the extended-reality gaming application is configured to modify the real and virtual objects in real or near real-time in response to the target location state data and display the modified real and virtual objects using the extended-reality device.

(64) In accordance with another aspect of the present invention, there is provided an extended-reality projectile-firing gaming system that includes a projectile-firing device comprising a projectile repository, a battlefield object, a network configured to provide communication and control connectivity in accordance with at least one protocol to the projectile-firing device and the battlefield object, an extended-reality gaming application configured to control the projectile-firing device and the battlefield object in accordance with a gaming session having a gameplay configuration for providing an extended-reality environment, receive projectile-firing device data from the projectile-firing device, the projectile-firing device data indicating that a projectile was fired, receive battlefield object data from the battlefield object, the battlefield object data indicating that an impact occurred on the battlefield object, and a computer-aided interface for interaction with gameplay control options associated with the gameplay configuration and continually updated

in real or near real-time during the gaming session, the computer-aided interface configured to receive commands or instructions associated with the real and virtual objects and corresponding to the gameplay configuration. The commands or instructions may be received using voice-activation.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The invention may be more readily understood by referring to the accompanying drawings in which:
- (2) FIG. 1 is a diagram of an extended-reality projectile-firing gaming system in accordance with a preferred embodiment of the present invention;
- (3) FIG. 2 is a diagram of a projectile-firing device in accordance with a preferred embodiment of the present invention;
- (4) FIG. 3A is a perspective view of a standalone target in accordance with a preferred embodiment of the present invention;
- (5) FIG. 3B is a perspective view of a monolithic target in accordance with a preferred embodiment of the present invention;
- (6) FIG. 4 is a perspective view of a battlefield environment depicting exemplary components of the extended-reality projectile-firing system in accordance with a preferred embodiment of the present invention;
- (7) FIG. 5 is a perspective view of a mixed-reality battlefield environment depicting exemplary components of the extended-reality projectile-firing system in accordance with a preferred embodiment of the present invention;
- (8) FIG. 6 is a top-down view of a battlefield environment depicting exemplary components of the extended-reality projectile-firing system in accordance with a preferred embodiment of the present invention;
- (9) FIG. 7 is a flow diagram of an extended-reality projectile-firing gaming method in accordance with a preferred embodiment of the present invention;
- (10) FIG. 8 is a flow diagram of an extended-reality projectile-firing gaming method in accordance with a preferred embodiment of the present invention.
- (11) FIG. 9 is a perspective view of one or more smart targets in accordance with a preferred embodiment of the present invention;
- (12) FIG. 10 is a perspective view of a wearable device including smart targets in accordance with a preferred embodiment of the present invention;
- (13) FIG. 11 is a smart blaster in accordance with a preferred embodiment of the present invention;
- (14) FIG. 12 is a schematic diagram of a circuit for a smart target co-witnessing hit attribution system in accordance with a preferred embodiment of the present invention;
- (15) FIG. 13-1 depicts a first portion of a schematic diagram of a circuit for a smart target co-witnessing hit attribution system in accordance with a preferred embodiment of the present invention;
- (16) FIG. 13-2 depicts a second portion of a schematic diagram of a circuit for a smart target co-witnessing hit attribution system in accordance with a preferred embodiment of the present invention;
- (17) FIG. 13-3 depicts a second portion of a schematic diagram of a circuit for a smart target co-witnessing hit attribution system in accordance with a preferred embodiment of the present invention;
- (18) FIG. 14A is a diagram of an exemplary smart target mesh network in accordance with a preferred embodiment of the present invention;
- (19) FIG. 14B is a diagram of an exemplary smart target mesh network in accordance with a

- preferred embodiment of the present invention;
- (20) FIG. 15A is a diagram of an exemplary smart target mesh network in accordance with a preferred embodiment of the present invention;
- (21) FIG. 15B is a diagram of an exemplary smart target mesh network in accordance with a preferred embodiment of the present invention;
- (22) FIG. 15C is a diagram of an exemplary smart target mesh network in accordance with a preferred embodiment of the present invention;
- (23) FIG. 16A is a diagram of an exemplary smart target mesh network in accordance with a preferred embodiment of the present invention;
- (24) FIG. 16B is a diagram of an exemplary smart target mesh network in accordance with a preferred embodiment of the present invention;
- (25) FIG. 16C is a diagram of an exemplary smart target mesh network in accordance with a preferred embodiment of the present invention;
- (26) FIG. 17A is a diagram of an exemplary smart target mesh network in accordance with a preferred embodiment of the present invention;
- (27) FIG. 17B is a diagram of an exemplary smart target mesh network in accordance with a preferred embodiment of the present invention;
- (28) FIG. 17C is a diagram of an exemplary smart target mesh network in accordance with a preferred embodiment of the present invention;
- (29) FIG. 18 is a table of a message protocol in accordance with a preferred embodiment of the present invention;
- (30) FIG. 19 is a table of smart target states in accordance with a preferred embodiment of the present invention;
- (31) FIG. 20 is a table of a message protocol in accordance with a preferred embodiment of the present invention;
- (32) FIG. 21A is a table of smart target states in accordance with a preferred embodiment of the present invention;
- (33) FIG. 21B is a table of smart target modes in accordance with a preferred embodiment of the present invention;
- (34) FIG. 22A is a first portion of a flow diagram of a smart target co-witnessing hit attribution method in accordance with a preferred embodiment of the present invention;
- (35) FIG. 22B is a second portion of a flow diagram of a smart target co-witnessing hit attribution method in accordance with a preferred embodiment of the present invention; and
- (36) FIG. 23 is a view of an extended-reality device in accordance with a preferred embodiment of the present invention.
- (37) Like numerals refer to like parts throughout the several views of the drawings.

#### DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(38) The following description and drawings are illustrative and are not to be construed as limiting. Numerous specific details are described to provide a thorough understanding of the disclosure. However, in certain instances, well-known or conventional details are not described in order to avoid obscuring the description. References to one or an embodiment in the present disclosure can be, but not necessarily are references to the same embodiment; and, such references mean at least one of the embodiments. If a component is not shown in a drawing then this provides support for a negative limitation in the claims stating that that component is “not” present. However, the above statement is not limiting and in another embodiment, the missing component can be included in a claimed embodiment.

(39) Reference in this specification to “one embodiment,” “an embodiment,” “a preferred embodiment” or any other phrase mentioning the word “embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure and also means that any particular feature, structure, or

characteristic described in connection with one embodiment can be included in any embodiment or can be omitted or excluded from any embodiment. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment, nor are separate or alternative embodiments mutually exclusive of other embodiments. Moreover, various features are described which may be exhibited by some embodiments and not by others and may be omitted from any embodiment. Furthermore, any particular feature, structure, or characteristic described herein may be optional. Similarly, various requirements are described which may be requirements for some embodiments but not other embodiments. Where appropriate any of the features discussed herein in relation to one aspect or embodiment of the invention may be applied to another aspect or embodiment of the invention. Similarly, where appropriate any of the features discussed herein in relation to one aspect or embodiment of the invention may be optional with respect to and/or omitted from that aspect or embodiment of the invention or any other aspect or embodiment of the invention discussed or disclosed herein.

(40) The terms used in this specification generally have their ordinary meanings in the art, within the context of the disclosure, and in the specific context where each term is used. Certain terms that are used to describe the disclosure are discussed below, or elsewhere in the specification, to provide additional guidance to the practitioner regarding the description of the disclosure. For convenience, certain terms may be highlighted, for example using italics and/or quotation marks: The use of highlighting has no influence on the scope and meaning of a term; the scope and meaning of a term is the same, in the same context, whether or not it is highlighted.

(41) It will be appreciated that the same thing can be said in more than one way. Consequently, alternative language and synonyms may be used for any one or more of the terms discussed herein. No special significance is to be placed upon whether or not a term is elaborated or discussed herein. Synonyms for certain terms are provided. A recital of one or more synonyms does not exclude the use of other synonyms. The use of examples anywhere in this specification including examples of any terms discussed herein is illustrative only, and is not intended to further limit the scope and meaning of the disclosure or of any exemplified term. Likewise, the disclosure is not limited to various embodiments given in this specification.

(42) Without intent to further limit the scope of the disclosure, examples of instruments, apparatus, methods and their related results according to the embodiments of the present disclosure are given below. Note that titles or subtitles may be used in the examples for convenience of a reader, which in no way should limit the scope of the disclosure. Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure pertains. In the case of conflict, the present document, including definitions, will control.

(43) It will be appreciated that terms such as “front,” “back,” “top,” “bottom,” “side,” “short,” “long,” “up,” “down,” “aft,” “forward,” “inboard,” “outboard” and “below” used herein are merely for ease of description and refer to the orientation of the components as shown in the figures. It should be understood that any orientation of the components described herein is within the scope of the present invention.

(44) Referring now to the drawings, which are for purposes of illustrating the present invention and not for purposes of limiting the same, the drawings show devices and components (and related methods) therein in accordance with preferred embodiments of an extended-reality projectile-firing gaming system and method. As shown in FIGS. 1-6, the extended-reality projectile-firing gaming system generally includes a system of wirelessly interconnected blasters, targets, wearables, and “battlefield” objects that allow players to engage in precisely tracked and managed projectile sports and games, resulting in an experience of “Video Games Outside”. As shown in FIGS. 7-8 extended-reality projectile-firing gaming methods are shown, further illustrating preferred methods for attributing scoring to particular players or users of the extended-reality projectile-firing gaming

system. As shown in FIGS. 9-23, a smart target co-witnessing hit attribution system and method are shown, illustrating particular embodiments of accurately determining “hits” and correctly attributing those hits to a particular projectile-firing device.

(45) Referring now to FIG. 1, FIG. 1 is a diagram of an extended-reality projectile-firing gaming system **100** in accordance with a preferred embodiment of the present invention. FIG. 1 depicts two users utilizing a network **102** in connection with the system **100**. The system **100** includes a projectile-firing device **104**, an extended-reality device **105**, a wearable device **106**, a standalone target **108**, a drone **109**, a cloud-based computing system **110**, a remote device **112**, and a fitness monitoring device **114**. The network protocol may include Bluetooth, WiFi, radio frequency (RF), or other suitable network protocol.

(46) The network **102** is configured to utilize one or more network protocols to facilitate communication by and between the various components of the system **100**. For example, as disclosed herein, the network **102** may consist of any network type, including but not limited to a local area network (LAN), wide area network (WAN), and/or the internet. One of ordinary skill in the art would understand that a variety of networks and/or network protocols may be implemented to provide connectivity between the various components of the system **100**.

(47) Referring to FIG. 1, two projectile-firing devices **104** are depicted. The first projectile-firing device **104** is associated with a first user **116** and the second projectile firing device **104** is associated with a second user **118**. One of ordinary skill in the art would understand that more than two devices **104** may be utilized and that the system **100** is scalable to include more than two users **116**, **118**. FIG. 2 is a diagram of a projectile-firing device **102** in accordance with a preferred embodiment of the present invention. The projectile-firing device **102**, in a preferred embodiment, includes a housing **120**, a handle **122**, a trigger **124**, a battery pack **126** (internal to the handle **122** in FIG. 2), a muzzle **128**, and a hopper **130**. The housing **120** preferably is configured to receive the handle **122**, the trigger **124**, the battery pack **126**, the muzzle **128**, and the hopper **130**. The housing **120** is also preferably configured to embody electromechanical apparatus to receive commands and/or instructions via the network **102**, feed projectiles from the hopper **130** to be fired by the projectile-firing device **104**, and mechanically fire one or more projectiles from the muzzle **128**. The electromechanical apparatus may include a gearbox configured to load and fire gellets, small biodegradable projectiles configured to be fired from the muzzle **128**. The handle **122** is configured to receive the trigger **124** and used by the user **116**, **118** to depress the trigger **124** to fire the projectiles from the muzzle **128**. When the trigger **124** is depressed by the user **116**, **118**, the device **104** may be configured to fire a single projectile, or multiple projectiles (e.g., automatic fire mode). In an embodiment, the device **104** includes a switch **133** to change the fire mode from single fire to automatic fire modes. Alternatively, or in combination with the switch **133**, the device **104** includes a display **152** (shown in FIG. 4) that is configured to change the fire mode from single fire to automatic fire modes. In an embodiment, the fire mode is configured to fire three projectiles upon the trigger being depressed once by the user **116**, **118**. The fire mode may also be controlled by the application via the network **102** to vary the fire mode depending upon the gaming configuration in accordance with a gaming session.

(48) The muzzle **128** is configured to permit the projectile from exiting the projectile-firing device **104**. The projectile-firing device, in a preferred embodiment, is configured to vary the velocity of a projectile from 90 feet-per-second (FPS) to 170 FPS. In other embodiments, the velocity is configured to reach 200 FPS or greater. For example, the projectile-firing device receives commands and/or instructions from the application via the network **102** to electromechanically adjust the velocity at which the projectile-firing device **104** is fired. The velocity may be adjusted electromechanically through a variety of methodologies, all within the scope of the present invention.

(49) In an embodiment, the electromechanical functionality of the projectile-firing device **104** includes a barrel **129** (shown internally with dashed lines) to be inserted into the muzzle **128** and/or

muzzle area. The barrel **129** may be configured to increase or decrease a velocity at which the projectile-firing device **104** fires a projectile. For example, a first barrel **129** is configured to achieve a muzzle velocity of 170 FPS, while another barrel **129** is configured to achieve a muzzle velocity of 90 FPS. Alternatively, or in combination with the barrel **129**, a barrel tip **131** is preferably configured to be affixed to the muzzle **128** and/or muzzle area. The barrel tip **131** also may be configured to achieve muzzle velocities within the velocity ranges described herein. In yet another embodiment, the barrel tip **131** may be configured to reduce the velocity from a predetermined “default” range; thus, for example, when the barrel tip **131** is not affixed to the muzzle **128**, the velocity is 170 FPS, while when the barrel tip **131** is affixed to the muzzle **128**, the velocity is 90 FPS. The barrel tip **131** may be configured to allow beginners or those desiring a less aggressive gameplay experience to lower the velocity. Still further, in any of the aforementioned embodiments, the mechanical components of the muzzle **128**, the barrel **129**, and the barrel tip **131** may be combined with electronically adjusted velocity increments so as to achieve FPS velocities greater or lower than the predetermined velocities of the mechanical components. Still further, the projectile-firing device **104** may include physically adjustable areas alongside the barrel **129** to open or close to vary the velocity mechanically. In an embodiment, the velocity is decreased when the physically adjustable areas are opened further, while the velocity is increased when the physically adjustable areas are closed further. The opening and closing of the physically adjustable areas may be controlled by the application via the network and/or internal electronic components of the projectile-firing device **104**.

(50) Other methods of lengthening or shortening ranges mechanically are within the scope of the present invention. In an embodiment, electronic adjustment of blaster spring tension may be utilized, meaning that certain weapon types can be made to fire physically longer or shorter distances as determined by the system, allowing for “range” as an attribute; e.g., Sniper vs Revolver (e.g., Table 1).

(51) The device **104** preferably includes electronic circuitry, or the like, to provide relatively accurate position and direction of the device **104**. This data preferably is utilized to provide confirmation, in connection with the application, of a “hit” by a particular user **116**, **118** on another user's **116**, **118** wearable device **106** or a particular standalone target **108**/monolithic target **148** or other target/battlefield object. The device **104** may also include an IR reader so that the system **100** in connection with the application can sense that a particular device **104** has reflected IR radiation on a wearable device **106** or target for further confirmation of a “hit.”

(52) Referring still to FIG. 2, the battery pack **126** is configured to provide power to device **104**. For example, the device **104** includes electromechanical apparatus such as internal electronic circuitry configured to be powered by the battery pack **126**. The battery pack **126** is also preferably configured to operate mechanical firing apparatus of the device **104**. The battery pack **126** preferably is a rechargeable battery pack **126** configured to receive a USB-C input to recharge the battery pack **126**. In other embodiments, the battery pack **126** may be recharged by a variety of recharge methods, all within the scope of the present invention, including a mini-USB input, or the like. The device **104** preferably includes a recharge port **127** configured to receive a recharge input.

(53) The hopper **130** preferably is configured to hold a “depot” of projectiles to be fed to the electromechanical apparatus within the device **104** for firing. In various embodiments, the hopper is configured to hold 100, 200, 300, 400, 500, 600, 700, 800, 1000, or greater or fewer projectiles. The gaming configuration in accordance with a gaming session may limit the number of projectiles held by the hopper **130**. The hopper **130** is configured to gravity feed projectiles into the device **104**.

(54) In a preferred embodiment, the projectile-firing device **104** is configured to uniquely receive instructions and/or commands from the application via the network **102** to increase or decrease electromechanical functionality of the projectile-firing device **104** when receiving “damage,” receiving points or “health,” or during a variety of other scenarios, preferably in connection with a



gaming configuration. The physical consequences of this capability impact the first user's **116** ability to inflict "damage" to the second user **118**. As a result, the rate of fire may be increased or decreased, a reload interval may be increased or decreased, a velocity of the device **104** may be increased or decreased, and an amount of "damage" inflicted can be varied depending on the commands and/or instructions, including the type and configuration of the gaming profile (e.g., gaming configuration) that is configured to be set during a gaming session. A projectile-firing device **104**, for example, may be set in a configuration as a sniper rifle, a pistol, or a shotgun. Each of these configurations, in a preferred embodiment, are preset with specified rates of fire, reload intervals, damage capabilities, range of fire, or the like. The projectile-firing device **104** set in sniper rifle mode, for example, would have a slower rate of fire, a more frequent reload interval, and a higher damage capability than the projectile-firing device set in pistol mode, for example. The projectile-firing device **104** is configured to be electromechanically controlled by the application via the network **102** such that any of the above configurable options may be varied in real- or near-real-time.

(55) The system **100** in accordance with the application gameplay configuration will allow or disallow the refill or usage of projectiles according to gameplay modes where a "reload" can be physically controlled from electromechanical assembly and connected via the network **102** and given instructions from the application. A projectile dispenser may be based on predetermined parameters within the gameplay and may embody or be part of the hopper **130**. For example, a game may be set where the users **116**, **118** (e.g., players) can only refill a certain number of times during the length of the game and after they have reached the limit the system prevents the player from reloading (e.g., locking the canister, stopping the blaster from working, etc.). The right to refill may be earned and the blaster may begin working again. In an embodiment, a home base/capture point station **172** may be utilized that permits the users **116**, **118** to refill projectiles literally (e.g., within the existing hopper **130**), as well as virtually per the game mode (e.g., gaming configuration).

(56) The projectile-firing device **104** is configured to monitor and track the time when a projectile is fired from the device **104**, and to provide this information to the application. As described herein, the system **100** is configured to receive the time the projectile was fired and correlate the time with the time a wearable device **106** was impacted by a projectile to determine that the user **116**, **118** associated with the wearable device was hit by the user **116**, **118** associated with a particular projectile-firing device **104**. This methodology achieves benefits above and beyond traditional projectile-firing game systems because it can monitor and track who is responsible for a hit, allowing for more interactive gameplay and immersion.

(57) The device **104** preferably includes a display configured to provide useful game information, such as hits, health, ammo, blaster "type", team alignment, remaining players, etc. The device **104** preferably includes electronic circuitry and a network interface to receive data from the network **102** in accordance with the application and the gaming configuration of that particular gaming session.

(58) Referring again to FIG. 1, two extended-reality devices **105** are depicted, separately worn by and associated with the users **116**, **118**. The extended-reality devices **105** are configured to communicate via the network **102** with the application. The extended-reality devices **105** may be augmented reality (AR), mixed-reality (MR), virtual reality (VR), or extended-reality (XR) devices configured to receive and display data via the network **102** to the users **116**, **118**. The extended-reality device **105** is configured to enhance game modes and game attributes by mixed and/or augmented reality. Any suitable head-mounted display, including those of Meta Quest, Nreal, or Apple Vision Pro, now and in the future may be utilized to display relevant game information in heads-up-display (HUD) style, allowing the users **116**, **118** to track gameplay configurations, statistics, etc., in a video-game-like manner. For example, the extended-reality device **105** includes a HUD display **142** configured to display current health, ammunition (i.e., projectiles),

weapon/blaster/projectile-firing device **104** type, shields, and the like. The HUD display **142** may also display “kills,” “deaths,” and a “kill/death” and/or success/failure ratio, including a ranking of users taking part in the gaming session. In a gaming session where certain of the users **116, 118** have been eliminated from the gaming session in accordance with the gameplay configuration, the remaining team members and opponents are displayed. In an embodiment, alerts, game start/end, timers, and special game states are displayed. The accuracy of the users **116, 118** may also be displayed as a shots/hits ratio. The HUD display **142** may also include display of the current objectives, game modes, progress, capture/flag/base status/health in accordance with the gameplay configuration. In certain embodiments, the gameplay configuration may utilize battlefield roles (e.g., medic, engineer, tank, sniper, etc.). These roles may include specialized perks and restrictions that may be displayed to the users **116, 118** via the HUD display **142**.

(59) Each of these displayable items may also utilize real-time or near-real-time virtual overlays on the battlefield environment displayed through the HUD display **142**. For example, health and shield bars, team affiliation, badges, status, role, username, etc. can appear over the users' **116, 118** heads. Battlefield objectives can be highlighted in the distance, and capture or base areas can be defined and shown virtually, object pickups can be described and highlighted, weapon skins can be virtualized and overlaid on the device **104**, and team colors, player skins, and other aesthetics can be mapped to human users **116, 118** in real-time or near-real-time. The HUD display **142** may also display zones of special status, such as neutral or safe, which can be virtually displayed on the battlefield environment, and hits, shield depletion, firing, etc. can be displayed as real-time or near-real-time virtual animations. Battlefield objects, skins, the environment, etc. can be themed to match game mode, user preference, and/or brand tie-ins, such as the Marvel Universe, WWII, or Sci-Fi themes. The application can restrict or unlock certain themes, nomenclatures, modes, etc., based upon player progress, preference, or parental controls. For example, realistic weapon terminology, violent descriptions such as “killed,” or “bloody” animation effects can be restricted to older players or turned off entirely.

(60) QR code/motion-capture-type patterns can be printed on blasters, targets, and battlefield objects to aid the extended-reality device **105** in overlaying virtualized animations/skins accurately utilizing the HUD display **142**.

(61) In accordance with various embodiments, the extended-reality projectile-firing gaming system **100** utilizes the extended-reality devices **105** to provide a selection interface configured to work within the system **100** ecosystem and control varied options within the game. In an embodiment, any of the selectable options described herein may be implemented through a selection interface viewable and usable through the extended-reality devices **105**. For example, the real and/or virtual objects described herein are associated with commands and instructions that are configured to be selected by the users **116, 118** using a computer-aided interface.

(62) The commands and instructions preferably include choosing the type of projectile-firing device **104** (e.g., type of blaster **104**, firing mode, etc.), choosing with whom to communicate during a gaming session, choosing map pullup and options, and the like. For example, controlling the blaster **104** mode may include toggling or selecting a single/multi-shot mode. In a preferred embodiment, the electromechanical apparatus may be controlled via the commands and instructions to fire in accordance with type/mode of firing, to fire faster and/or harder, or the like. A single-fire revolver example may include a single-shot mode of firing, a relatively slower/lighter projectile speed, and a specified number of reloads (e.g., six). A multi-shot rifle, on the other hand, may include an automatic/burst mode of firing, and may speed up the projectile speed with a higher number of reloads. These different blaster **104** types are configured to be selectable using the computer-aided interface in association with the extended-reality devices **105**. In addition, the commands and instructions preferably include controlling rate of fire, such as 8 projectiles per second versus 12 projectiles per second.

(63) Besides blaster **105** control, other controllable options include other in-game implement/tool

selection, ammunition selection, map review and expansion, equipping of different accessories, and communication. In an embodiment, the controllable options may include a power-up menu selection, such as potions or other items found in the extended-reality environment world. For example, when the user's **116** health is low, a virtual potion may be selected and used to increase the user's hit points. Other examples are invincibility shields, increased speed, or the like. Map review and expansion preferably uses the location of other headsets (e.g., extended-reality devices **105**), blasters **104**, or other real object for which location services are available (e.g., remote device **112**) for locations of other players within the map. The different accessories may include lights, tracer rounds, and night vision, among other like accessories. For example, these accessories may be real or virtual items, or may simulate real-world items. Night vision goggles, for example, may be simulated by providing a green “glow” through the extended-reality device **105**. A light, on the other hand, may be a real-world object that illuminates the gaming environment, or may be an effect that lightens the extended-reality environment through the device **105**.

(64) The controllable options also include communication among teammates or other users **116**, **118**. For example, the user **116** sees a dropdown menu of teammates through the extended-reality device **105** and selects the one (or ones) that the user **116** would like to speak with. The computer-aided interface facilitates selection of communication options, including selection of a particular user with whom to communicate. For example, eye contact with a map icon depicting a user **116** for a long enough predetermined time period may be sufficient to open up communications with that user **116**.

(65) In an embodiment, the extended-reality devices **105** include the computer-aided interface, but in other embodiments, the interface is separate from the extended reality devices **105**. For example, the extended-reality devices **105** include one or more eye sensors configured to monitor eye contact of the users **116**, **118**. The eye sensors may be accompanied by an algorithm designed to monitor how long the users' **116**, **118** eyes are maintaining eye contact with a particular real or virtual object displayed by the extended-reality devices **105**. In an embodiment, the algorithm compares the time of eye contact with a predetermined time period (e.g., 1 or 2 seconds) to confirm a selection of the particular real or virtual object (e.g., a command or instruction associated with the real or virtual object). The algorithm may include a range such that eye contact greater than a first predetermined time period and less than a second predetermined time period confirms a selection.

(66) However, the algorithm may simply monitor eye contact without requiring the eye contact to be maintained within a certain time period. For example, if the user **116**, **118** makes eye contact with a real or virtual object, the user **116**, **118** may activate a command or instruction associated with the real or virtual object by using voice activation and/or selection by hand movement or remote device, or any combination thereof. For example, the user **116** makes eye contact with a “Revolver” blaster choice and says “Select” to choose the Revolver as the blaster **104** type. Or, the user **116** makes eye contact with a “Revolver” blaster choice and depresses a button on the blaster **105** to choose the Revolver as the blaster **104** type. For example, the button on the blaster **104** may be selectable via the remote device **112** that is affixed to the blaster **104** or may be separately provided from the blaster **104**. In other embodiments, the computer-aided interface may consist of separately implemented eye sensors, hand movements, physical controllers, remote devices **112**, or combinations thereof. For example, a trackpad, ball mouse, or similar physical system may be utilized to control a menu within a game. The physical controller can consists of merely up/down/left/right buttons on the blaster itself, or may be a separate standalone controller. The hand movement may be recognizable by use of one or more sensors using any of the sensor types described herein, such as LIDAR, cameras, or the like.

(67) In an embodiment, the blaster **104** control via the computer-aided interface may be integrated with hit detection/attribution methods as disclosed herein. For example, the methods disclosed herein may output updated gaming metric data and/or target location state data. In response to that data, the real and virtual objects may be modified to reflect real or near real-time events. The

modified real and virtual objects may then provide updated commands or instructions available to be selected by the users **116**, **118**. For example, a user **116** may confirm a hit using the methods provided herein that, when updated via the updated gaming metric data and/or target location state data, triggers an in-game response, such as providing power-ups or additional in-game capabilities. The user **116** is then able to select the in-game virtual (or real) item as it appears or is prompted to be utilized according to the computer-aided interface of the extended-reality device **105**.

(68) In an embodiment, the extended-reality device **105** utilizes the computer-aided interface to prompt the users **116**, **118** to select or input a command or instruction.

(69) The computer-aided interface of the extended-reality device **105** is designed to minimize or eliminate negative impacts on gameplay or the users' **116**, **118** experience and provide an immersive gameplay environment.

(70) In an embodiment, the extended-reality gaming system **100** utilizes audio to provide the computer-aided interface control options as described herein. In this embodiment, the system **100** does not rely on display of the real and virtual objects through the extended-reality device **105** to provide selectable options, and may not be needed at all. In this embodiment, the computer-aided interface utilizes verbal commands. For example, the system **100** includes microphones and earbuds or headphones (or relies upon external speakers and microphones affixed to the blaster **104**, wearable vest **108**, and/or remote device **112**) to provide the computer-aided interface for commands and instructions. For example, the user **116** states "Computer: identify blaster inventory" into a microphone, and the extended-reality gaming application responds through the speaker "Revolver, sniper rifle." The user **116** may then state into the microphone "Computer: Select revolver," and the blaster **104** becomes the revolver in that gameplay session. As described herein, the various audio options may be integrated with the extended-reality device **105** such that all or portions of the audio control options may be utilized alongside or in connection with the display of the real and virtual object and associated commands and instructions.

(71) Referring now to FIG. **23**, there is depicted a blaster **104** as displayed in an extended-reality device **104**. For example, the device **104** is a pair of MR goggles **104**. The MR goggles **104** display both real and virtual objects. For example, the floor tiles **580**, the projectile **164**, and the barrier **178** are all real objects. However, the ceiling tiles **582** are virtual objects. In addition, there is depicted in FIG. **23** a blaster choice dropdown menu **584** and a communication dropdown menu **586**. These are also virtual objects within the extended-reality environment. The computer-aided interface facilitates interaction with any of the real or virtual objects, including those depicted in FIG. **23**. One of ordinary skill in the art would understand that the real or virtual objects depicted in FIG. **23** are non-limiting. In practice, in an embodiment, the user **116** makes eye contact with "Blaster Choice", which drops down three options: Revolver, Shot Gun, and Sniper Rifle. The user **116** then selects one by making eye contact with Shot Gun, for example, and using a selection method, as disclosed herein, to select Revolver. The computer-aided interface receives the command/instruction to select Revolver and provides it to the application, which is then used to control the gameplay characteristics. The selection of Revolver preferably selects limitations and requirements that are communicated to the blaster **104** or the application to appropriately limit the gameplay in connection with the selection criteria. One of ordinary skill in the art would understand that a variety of methodologies of selecting, processing, and implementing the selections are within the scope of the present invention.

(72) Referring again to FIG. **1**, two wearable devices **106** are depicted. One of ordinary skill in the art would understand that while the wearable devices **106** are depicted as wearable vests, other wearables or body adornments affixed to other parts of the body may be utilized without departing from the scope of the present invention. The wearable devices **106** can have lighting profiles, sounds, and other controlled actions to integrate with gameplay. The first wearable device **106** is associated with the first user **116** and the second wearable device is associated with the second user **118**. The wearable device **106** includes a torso portion **132** and a shoulder portion **134**. As shown in

FIG. 1, the torso portion **132** and the shoulder portions **134** are attached via straps **136**. The torso portion **132** and the shoulder portions **134** of the wearable device **106** include wearable device targets **137** configured to detect impacts of projectiles on the wearable device **106**. The wearable device targets **136** are in communication with one or more of the device **104**, the standalone target **108**, the drone **109**, the cloud-based computing system **110**, the remote device **112**, and the fitness monitoring device **114**. Each of these components is configured to communicate with one or more of the components of the system **100** via the network **102**.

(73) Upon detection of a “hit,” the wearable device **106** is configured to send a message via the network **102** to the application indicating that the user **116**, **118** associated with the wearable device **106** has been hit by a projectile. As mentioned herein, when the application receives the message that the wearable device **106** has been hit, the application correlates the time the projectile was impacted on the wearable device **106** with a time a projectile was fired by a projectile-firing device **104**. The user **116**, **118** associated with the projectile-firing device **104** that was correlated with the impact on the wearable device **106** is then credited with a hit against the user **116**, **118** associated with the wearable device **106**.

(74) The standalone target **108** is depicted as a “flower” arrangement in a preferred embodiment. The standalone target **108** includes a central portion **138** (e.g., “pistil” of flower) and a plurality of outer portions **140** (e.g., “petals” of flower). As shown in FIG. 1, the standalone target **108** includes six outer portions **140**, but in other embodiments, the number of outer portions **140** may be greater or fewer than six. Each of the central portion **138** and the outer portions **140** are configured to separately determine whether a projectile has impacted that portion of the standalone target **108**. For example, the same target impact methodology is utilized for the wearable device targets **136** and the central and outer portions **138**, **140** of the standalone target **108**. In accordance with the gameplay configuration, “gamifying” the standalone target **108** may include a requirement to separately “hit” each of the portions **138**, **140** in order to credit a gameplay objective. In other embodiments, one or more of the central portion **138** and outer portions **140** are required to be hit in accordance with the gameplay configuration. The standalone target **108** permits a variety of gameplay options and/or objectives, all within the scope of the present invention.

(75) The standalone target **108** is preferably in communication with the application via the network **102** so that real-time or near-real-time data may be provided to the application, and visa versa. For example, upon a hit of the central portion **138** of the standalone target **108**, the application may credit a “hit” using similar methodology as that used to credit a “hit” on the wearable device **106**. The standalone target **108** may also be configured to include a physical indication (i.e., a red lighted portion) to indicate a hit upon a certain portion **138**, **140** of the standalone target **108**. In addition, or alternatively, the application, upon receiving data from the network **102**, may be configured to provide a virtual overlay on the target displayed on the display of the extended-reality device **105**.

(76) Referring now to FIGS. 3A and 3B, the standalone target **108** in FIG. 3A is shown in the same configuration as that shown in FIG. 1. The standalone target **108** includes support members **144** to support the central portion **138** and the outer portions **140**. The standalone target **108** also includes a stand **146** configured to support the standalone target **108** in an upright position. As shown in FIG. 3B, a monolithic target **148** is shown, which is an alternative configuration of the standalone target **108**. The monolithic target **148** includes a similar arrangement including a central portion **138** and outer portions **140**, each configured to determine whether a projectile has impacted that particular portion of the monolithic target **148**. The monolithic target **148** is constructed so that the central portion **138** and the outer portions **140** are embedded in the same structure (i.e., monolithic construction) or mold **150**. The monolithic target **148** may be affixed to a structure such as a barrier or battlefield object, or may be positioned in an upright position using a stand, similar to that shown in FIG. 3A. Similarly to the standalone target **108**, the monolithic target **148** is preferably configured to communicate with the application via the network **102**, and includes the same “hit”

confirmation methodology as that described herein in connection with the standalone target **108** and the wearable device **106**.

(77) While specific seven-panel-shaped targets (e.g., standalone target **108**, monolithic target **148**) have been shown, other shapes or configurations are within the scope of the present invention.

(78) The system **100** as shown in FIG. **1** includes a drone **109**. The drone **109** is configured to communicate via the network **102** with the application. The drone **109** may be preprogrammed to provide a “passive” target similar to the standalone target **108** or the monolithic target **148**, and/or may be an “active” target or participant preprogrammed to be controlled remotely. In an embodiment, the drone **109** includes a projectile-firing device **104** affixed thereto. One of ordinary skill in the art would understand that the particular projectile-firing device **104** depicted in FIG. **1** would be modified to be affixed to the drone **109**.

(79) Referring still to FIG. **1**, the cloud-based computing system **110** is configured to provide on-demand availability of computer system resources. The cloud-based computing system **110** preferably includes data storage and computing processor resources to offload storage and processing into a cloud environment. One of ordinary skill in the art would be familiar with cloud-based computing systems and would understand that a variety of configurations of components are within the scope of the present invention. The cloud-based computing system **110** is preferably configured to store and run the application.

(80) The remote device **112** as depicted in FIG. **1** is preferably a smartphone and includes direction-sensing and location-sensing capabilities. For example, the remote device **112** includes GPS and a magnetometer to determine position and direction of the device **112**. As disclosed herein, the remote device **112** may be affixed to the projection-firing device **104** to provide relatively accurate position and direction of the device **104**. This data preferably is utilized to provide confirmation, in connection with the application, of a “hit” by a particular user **116**, **118** on another user's **116**, **118** wearable device **106** or a particular standalone target **108**/monolithic target **148** or other target/battlefield object. The remote device **112** may also include an infrared (IR) reader so that the system **100** in connection with the application can sense that a particular device **104** has reflected IR radiation on a wearable device **106** or target (e.g., the targets **106**, **148**, the drone **109**) for further confirmation of a “hit.”

(81) The remote device **112** is preferably connected to the network **102** and configured to communicate with the application. The remote device **112** in a preferred embodiment is configured to receive real- or near-real-time data from the network **102** to provide commands and/or instructions to the user **116**, **118**, and to display the commands and/or instructions thereon in accordance with the gaming configuration for that particular gaming session. The remote device **112** may display useful game information, such as hits, health, ammo, blaster “type”, team alignment, remaining players, etc.

(82) The fitness monitoring device **114** as depicted in FIG. **1** is configured to communicate with the application via the network **102**. The device **114** may be a smart watch such as Apple Watch and Samsung Gear for integration with additional mobile applications. For example, the device **114** may be configured to integrate with health apps and features via an API. In an embodiment, the users **116**, **118** may gain XP or other in-game features through increased physical activity and exertion. Users **116**, **118** that exhibit more active gameplay, monitored through the fitness monitoring device **114**, may earn increased XP. The users **116**, **118** may use the XP to unlock special items, exclusive products, free items, or the like.

(83) As mentioned herein, the application is configured to be stored and run on the cloud-based computing system **110**. For example, the application is configured to receive, via the network **102**, data from all physical components of the system **100** and stored on a database associated with the cloud-based computing system **110**. The application may be configured to receive data from one or more wireless protocols such as Bluetooth (e.g., Bluetooth Low Energy (BLE)), WiFi, RF, etc.). The application preferably is configured to allow the users **116**, **118** to download new game modes,

receive commands and/or instructions, display score and skill-improvement tracking, and the like. The application may be configured to communicate with the remote device **112** to receive and display this information to the users **116**, **118**. The application is also configured to provide historical, up-to-date information concerning global and/or local scoreboards, such as those provided in video gaming environments.

(84) The application is also configured, in an embodiment, to provide control over light and/or sound preferences, customized gaming mode/gaming configuration creation or sharing, and the like.

(85) The application, in a preferred embodiment, is also configured to control and communicate with the various targets and physical objects in the system **100** via the network **102**. For example, the application is configured to control the start/stop of a gaming session, provide live, real- or near-real-time scoring updates, and visual/audible/haptic user feedback in real- or near-real-time. In accordance with a gaming configuration, the application configures the various components of the system **100** initially in a gaming session setup. Each of the “nodes” within the system **100** is configured to communicate with nearby “nodes” such that the application need only be connected via the network **102** to one other component within the system **100**. Thus, sharing and communication of data, information, commands and/or instructions, and the like may be shared between devices, and not necessarily through a “central” hub or routing station.

(86) Referring now to FIG. **4**, a battlefield environment depicting exemplary components of the extended-reality projectile-firing system **100** is shown. The battlefield environment depicted in FIG. **4** is an outside arena setting showing multiple users **116**, **118** each with a projectile-firing device **152**. The device **152** is the same or substantially the same as that depicted in FIGS. **1** and **2**, but includes a display **154** on the housing **120** facing the user **116**, **118**. One of ordinary skill in the art would understand that the display **154** may be placed on other areas of the housing **120** without departing from the scope of the present invention. The display **154** is configured to provide useful game information, such as hits, health, ammo, blaster “type” (e.g., projectile-firing device **104**, **152**), team alignment, remaining players, etc. For example, a health/shield meter **156** depicted in FIG. **4** shows that the user **116** has 100% of the available health points remaining, and a fully available shield. For example, the inner ring depicted on the health/shield meter **156** may signify the remaining health, while the outer ring depicted on the health/shield meter **156** may signify the remaining shield. The upper left corner of the display **154** shows team information **158** including a team affiliation, players on the team, and opponent players. The team information **158** may also indicate that one or more players have been eliminated from game play. The display **154** also includes, on the lower right corner, device information **160**, which includes the current weapon type (e.g., SURGE), the firing mode (e.g., semi-automatic, full automatic), the range (e.g., RNG: **4**), the damage capacity per round (e.g., DMG: **5**), and the number of rounds remaining (e.g., nine rounds). The range, for example, may be determined by a level system wherein a “default” blaster may be given a Level 1 rating. For example, the Level 1 specifications for the blaster may be represented by a single star next to the type of blaster, and include RNG: **4** (out of **10**), DMG: **5** (out of **20**), set to semi-automatic only, and 9 rounds available per reload. The range may be a limiting range, such that once a blaster fires at a target or user beyond that range, the “hit” will not be credited toward the blaster's user. Alternatively, the range may be a dampening factor such that beyond the optimum range the blaster's damage capabilities is depleted significantly. Through the networked system **100**, the application can determine the range at which a device **152** was fired at a target **108**, **148** or a user **116**, **118**, and attribute damage to the user associated with that particular device **152** based on the distance between the device **152** and the target **108**, **148** or the user **116**, **118**.

(87) Through gameplay, upgrades to the blaster level can be acquired in addition to other blaster types. Upgrades may increase the range, damage, and rounds available of a given blaster type. Other blaster types such as “shotguns” could be picked up, and would have specifications balanced

differently, such as very high damage ceilings, reduced range ceilings, and varying round capacities.

(88) The display further includes, on the upper right corner, power/connectivity information **162**, which includes icons representing, for example, WiFi and Bluetooth signals, remaining device power, etc. In this embodiment, both the WiFi and Bluetooth protocols are utilized to provide the network **102**.

(89) The device **152** depicted in FIG. **4** also shows a hopper **130** with projectiles **164** therein. As described herein, the projectiles **164** may be gellets, paintballs, or any other type of projectile suitable for gameplay. The users **116**, **118** depicted in FIG. **4** are shown wearing glasses **166**. The glasses **166** may be simple safety glasses, or in an embodiment, may be extended-reality devices **105**.

(90) Referring now to FIG. **5**, a mixed-reality battlefield environment depicting exemplary components of the extended-reality projectile-firing system **100** is shown. FIG. **5** depicts a first-person view looking through an extended-reality device **105**. As shown, the extended-reality device **105** includes the HUD display **142** through which both real-world physical objects such as the users **116**, **118**, trees, and ground are viewable, as well as virtual objects such as the health/shield meter **156**, the team information **158**, the device information **160**, and the power/connectivity information **162**. FIG. **5** also includes player information **168** displayed in a pop-up style banner above the user **116**, **118**. For example, as shown in FIG. **5**, the player information **168** identifies the name of the player (e.g., JESSTHETIGER), the player's affiliation (e.g., shown by the icon to the left of the player's name), and the player's health/shield meter **156** within the pop-up banner. FIG. **5** further includes an in-game message **170** configured to provide the user **116**, **118** associated with the extended-reality device **105** information about the gaming session. For example, as shown in FIG. **5**, the user **116** identified as "CASEYTHELION" is shown to be "KNOCKED OUT!", indicating that the user **116** has been eliminated from gameplay. Other information and/or data, as described herein, may be displayed to the users **116**, **118** through the HUD display **142** without departing from the scope of the present invention.

(91) Referring now to FIG. **6**, a top-down view of a battlefield environment in accordance with an exemplary extended-reality projectile-firing system **100** is shown. The various components depicted in the battlefield environment of FIG. **6** include projectile-firing devices **104**, **152** (e.g., blasters), standalone targets **108** (or monolithic targets **148**), a drone **109**, home base/capture point stations **172**, bunkers **174**, turrets **176**, and barriers **178**.

(92) As described herein, the projectile-firing devices **104**, **152** are each associated with a user **116**, **118** (not shown in FIG. **6**). The user **116**, **118** preferably is associated with an extended-reality device **105**, a wearable device **106** (e.g., wearable vest), a remote device **112** (e.g., a smartphone), and a fitness monitoring device **114**. In addition to these components, a user **116**, **118** may utilize physical player-borne shields, designed to shield or deflect projectiles **164**. These could be provided to certain "roles" (e.g., Tank, Heavy), or picked up on the battlefield. An iteration of this device could have a collapse/expand function, so that it only activates under certain game conditions, or for certain time intervals with a recharge period, but can be equipped during the whole game.

(93) Each of these components are capable of being in communication with the application via the network **102**, as described more fully herein. Not all components associated with the user **116**, **118** must be operatively connected to a "central" routing station, but may communicate data with nearby elements, associated with the user **116**, **118**, another user **116**, **118**, or any of the networked components described herein. For example, the standalone targets **108**, the drone **109**, the home base/capture point stations **172**, the bunkers **174**, and the turrets **176** are all configured to be in communication with the network **102**.

(94) The battlefield environment of FIG. **6** depicts two teams or "sides", each comprising three users **116**, **118** (e.g., players). The first team comprises three players on the eastern (right) side of



the battlefield, while the second team comprises three players on the western (left) side of the battlefield. On the first team's side of the battlefield, a standalone target **108** is associated with the first team. Thus, in an embodiment, one objective of the second team is to eliminate the standalone target **108** on the first team's side. For example, the target **108** may require a team to inflict damage or cause a certain number of hits to win, to unlock bonuses, “debuff” the enemy team, or as a target training game mode. Debuffing an opposing team may include reducing the team's rate of fire, reload times, number of health points, reduce shields, or a variety of other reduction methodologies. The debuff may be configured to be on a timer, such that when the timer expires, the debuff is removed.

(95) In addition, the first team's side includes a drone **109** (which may or may not be associated, or on the same team, as the first team; i.e., neutral, or otherwise), a home base/capture point station **172**, two bunkers **174**, a turret **176**, and two barriers **178**. The second team's side includes a standalone target **108**, a home base/capture point station **172**, a bunker **174**, two turrets **176**, and two barriers **178**. All of these battlefield components may be networked via the network **102** and configured and controlled by the application.

(96) A battlefield objective may include “capturing” the drone **109** to bring it onto the first team's side or the second team's side. The drone **109** may include a projectile-firing device **104** mounted and independently remotely controlled by an operator or user **116**, **118**. The drone **109** may include a kit enabling attachment of a projectile-firing device **104**, a standalone target **108**, or the like, to an existing drone platform. In an embodiment, the drone **109** may be integrated with these components and provided as an integrated unit.

(97) If risk of physical damage to the drone **109** is a factor, actual projectile firing could be disabled when the projectile-firing device **104** (e.g., a blaster held by a user **116**) is aimed at the drone **109**, counting as being “shot down” when the trigger is depressed while aimed at the drone **109**. The drones **109** may be configured to be “unlocked” as a perk for accomplishing certain objectives. The drones **109** may have certain automated features, allowing them to fly patterns remotely, such as hovering over the home base/capture point station **172** acting as defense, or following a given player, such as in certain video games. The drone **109** may be configured to execute “air strikes,” to be unlocked and called in, where the drone **109** takes off automatically and goes to a point or target, takes an action such as blasting it, and then flies back to a preset location.

(98) While the drone **109** is shown in FIG. 6, other mobile platforms such as rovers or other motorized objects may be utilized for enhanced gameplay. For example, the motorized objects may implement navigational controls, live streaming video, control of projectile-firing devices **104** installed onto the motorized objects, and hit detection and correlation in connection with standalone targets affixed to the motorized objects.

(99) In an embodiment, the motorized objects are ground-based remote controlled vehicles with mounted blasters (e.g., projectile-firing devices **104**) or “explosives” (e.g., mines, grenades, etc.), which can be activated by the user **116**, **118** or a remote operator. Both rovers and drones may be used as intelligence gathering with or without mounted blasters, where a camera feed is shown on the display **154** of a device **152**, the HUD display **142**, remote device **110**, or other viewable display. Control over these units could be contested as part of gameplay, where there is only one available rover (or drone **109**), and certain successful actions give one player or team access to it for a period of time.

(100) The home base/capture point stations **172** are configured to communicate with the application via the network **102**. The stations **172** allows players to refill projectiles (e.g., ammunition or rounds) physically, as well as digitally, per the gaming mode or configuration. Proximity to the stations **172** can offer “healing,” restoring lost health points while in-game. The stations **172** may also serve as a respawn point; for example, if a player is eliminated during gameplay, the player must physically return to the player's team station **172** in order to return to the game. The game mode or configuration may determine the respawn timer (i.e., how long a player

must wait to respawn), as well as the number of respawns possible per player. In addition, these stations **172** can serve as extensible capture points in appropriate game modes, where one team must hold multiple points simultaneously by positioning their players near it and keeping the opposing players away. Proximity can be determined by RF, NFC, GPS, blasting the object within a repeating time window, or a combination of these and/or other methods.

(101) The bunkers **174** in an embodiment are rapidly inflating and self-storing inflatable fold-out panels composed of rubber sheets, bistable spring sheets, and inflatable structural beams. The bunkers **174** include an attached pump or compressed air storage tank whereby the inflatable structural beams are inflated and the air pressure folds out the panels. Fixtures at the base of the unit can assist in maintaining the panel firmly in place. Inflation and deflation of the bunkers **174** may be controlled by the application and/or the users **116**, **118** utilizing any of the methods described herein to activate or deactivate battlefield objects. For example, the users **116**, **118** may deploy the bunkers **174** by activating the bunkers **174** via the remote device **112**, the display **154**, or through an interface available through the extended-reality device **105** and/or the HUD display **142**. The bunkers **174** may also be manually inflated/deflated by the users **116**, **118**. The application may also keep track of which bunkers **174** are deployed/inflated, and their location or proximity to other battlefield objects in accordance with several embodiments disclosed herein.

(102) The turrets **176** are configured to be in communication with the network **102** and controllable through a variety of methods, as disclosed herein. For example, global, connected true “video games” can be augmented into the application to allow for control of a turret **176** via the internet from users worldwide. The turret **176** may be controllable via the application by a team's user **116**. Controlling the turret **176** may include placing the turret **176** in a “sentry” mode, where if an opposing player is within a range of the turret **176**, the turret **176** will fire projectiles **164** at the opposing player. The turret **176** may include motion sensors or body-detecting cameras to allow for “sentry” functionality. The turrets **176** may be configured to be semi-autonomous (e.g., partially player controlled) and fully autonomous (e.g., player puts turret **176** into “mode”). The turrets **176** preferably include a standalone target **108** so that hit detection and correlation may occur. This functionality would allow a team to “take out” or deactivate a turret **176** during gameplay, or capture the turret **176** for the opposing team's advantage. The ability to detect hits and correlate the hits with particular players in the battlefield scenario may enhance gameplay with rewards, loot, or other asymmetrical advantages to players for completing interactions according to the active gameplay mode or configuration.

(103) In an embodiment, according to an “adversarial” target mode, a device **104** mounted on or near the target is aimed at the user **116**. Under certain game modes, the turret **176** would fire back at the player (e.g., a “wild west quick draw” mode). If the user **116** does not hit the target quickly enough, the turret **176** shoots back and the user **116** loses the round. This same mode could also be configured to operate with a standalone target **108**. The adversarial mode can also be configured to enable a human player to remotely control the adversarial blaster via internet through the network **102**.

(104) As with the bunkers **174**, the turrets **176** may be player-deployable or “permanent” (e.g., at a venue) turrets **176** that can be controlled directly or remotely, and/or be automated via motion detection, computer vision, or at random. The turrets **176** preferably would contain the same hit detection systems as the wearable devices **106**, the standalone targets **108**, and the monolithic targets **148**, allowing players to “destroy” them by hitting them a number of times, and allowing attribution of the turrets' **176** hits to the appropriate player or team. These turrets **176** could allow disabled or remote players to engage with their friends on the battlefield, either via the internet, nearby RF, or hands-on controls.

(105) The barriers **178** preferably are deployable static barriers, similar to the bunkers **174**, but “permanent” for that particular gaming session. For example, teams could be given a number of player-portable barriers to set up before the match, enabling a more interesting playspace on an

otherwise flat arena such as a sports field. The barriers **178** can be constructed of lightweight but durable fabric on a collapsible frame similar to a folding chair or umbrella. They can also be printed with branded or themed livery.

(106) As described herein, global, connected true “video games” can be augmented into the application to allow for control via the internet from users worldwide. Semi-autonomous or fully autonomous extensions may include “swarms” or “strikes” from mobile platforms for tournaments or larger-scale battles.

(107) In place of the permanent or semi-permanent battlefield objects illustrated in FIG. 6, capturable objects may be deployed in accordance with certain game modes. For example, in “capture-the-flag” gaming modes, the capturable objects preferably can detect their proximity to a team's home base/capture point station **172** through RF or NFC, or the like, and know when they are in motion via accelerometer and/or GPS, allowing the network to alert players when a “flag” is being captured.

(108) Beyond capturable objects, the battlefield objects illustrated in FIG. 6 may include battlefield “ordinance” and accessories. For example, the battlefield objects may include mines that are deployable, proximity detonated battlefield objects that blast projectiles outward. In an embodiment, the mines do not blast projectiles outward, but are controlled by the application to credit a “hit” from the mine in connection with a particular user **116** within range of the mine proximity. In this case, a player coming too close to a mine is notified via audio/video/haptic feedback that they have been hit, and the hit credited against the player. This can also work without projectiles, where a player coming too close is simply notified via A/V/H feedback that they have been hit. Alternately, the projectiles can be ejected, outward but not be required to impact the player-worn target in order to count, and thus simply providing tactile feedback for the hit.

(109) The battlefield objects may also include thrown objects by the player. Similar to the mines described above, “grenades” may be utilized and would simulate an explosion.

(110) As described herein, the type of weapon or projectile-firing device **104** may be virtual, utilizing the network **102** and NFC or other electronic means to allow for battlefield pickup of different weapon types or accessories. For example, a small object preferably can be picked up and placed near or attached to the Smart Blaster (or via interaction with a static battlefield object, e.g., shooting a target), telling it to change configuration and registering it in the system as a new weapon type. Table 1 below illustrates potential examples of various weapon types within the scope of the present invention in accordance with a gaming mode or configuration.

(111) TABLE-US-00001

| TABLE 1   | Type      | Capacity   | Reload Time | DMG per hit     | Rate of Fire |
|-----------|-----------|------------|-------------|-----------------|--------------|
| Revolver  | 6 rounds  | 3 seconds  | 1           | 1 shot per sec. |              |
| Automatic | 30 rounds | 5 seconds  | 2           | physical        |              |
| Bazooka   | 1 round   | 10 seconds | 20 (splash) | N/A             |              |
| Sniper    | 1 round   | 4 seconds  | 7           | N/A             |              |

(112) The “Revolver” of Table 1 includes a six-round capacity, requiring a player to reload the Revolver after firing six shots. To reload another six rounds, the Revolver is controlled to wait 3 seconds before being able to fire another round from the next six rounds. Each round inflicts 1 DMG per successful hit. The rate of fire is limited to 1 shot per second. The “Automatic” blaster of Table 1 includes a 30-round capacity with a reload time of 5 seconds. The DMG per hit for the Automatic is higher than the Revolver, at 2 DMG per hit. The rate of fire is not limited and may utilize the physical rate of fire available to the projectile-firing device **104**. Alternatively, the rate of fire may be limited. The Bazooka of Table 1 is a “splash-damage” capable option. Since the Bazooka is a slower-firing weapon type, and inflicts greater damage per hit, its reload time is slower than other types of weapons and does not have more than one round in its capacity. The Sniper of Table 1 includes a 1 round capacity with a 4-second reload time, which limits the rate of fire by a 4-second interval. The Sniper has a greater damage capacity of 7 damage than the Revolver or the Automatic. One of ordinary skill in the art would understand that the various weapon types and parameters are fluid and may change depending on the gameplay configuration. For example, a range or “overheat” timer could be specified by weapon type.

(113) In an embodiment, the trigger **124** of the device **104** is electronic; thus, the “reload” and related reload time can be enforced by disabling the trigger until the player hits a reload button, after which a reload timer is activated. Audible, visual, and haptic feedback can be provided for all of these actions and states (e.g., out of “digital” rounds/ammunition).

(114) Similarly, the rate of fire can be controlled, only registering a trigger **124** pull or firing a projectile **164** at the appropriate time interval. Semi and fully automatic modes may be manually or automatically controlled, as described herein.

(115) The system **100** calculates damage digitally through data received from the network **102**. For example, the device **104** provides a signal indicating its weapon type; if the target **108**, **148** or wearable device **106** is “hit,” the target **108**, **148** or wearable device **106** will know the weapon type and use the weapon characteristics (e.g., as disclosed in Table 1) to calculate the damage. Alternatively, or in combination, the application calculates damage inflicted on a battlefield object by a particular weapon type. The application, hosted in the cloud-computing system **110**, may store a lookup table or other suitable tracking method within a database, or the like, to track and credit hits by/to certain battlefield objects, as more fully described herein. In an embodiment, the damage inflicted by a particular weapon type may be degraded or decreased at outer limits of the weapon type's range.

(116) Splash damage, such as from a Bazooka, mine, or grenade, can be calculated by a proximity system, as more fully described herein. The splash damage may be greater the closer the battlefield object/target **108**, **148** or wearable device **106** is from the epicenter of the “explosion.” An IR device associated with the projectile-firing device **104** may be utilized to track splash damage. For example, a number of targets may be “pre-painted” with IR, which will hold them as painted for a fixed period of time, and if one is hit with a projectile within that window, they each receive damage.

(117) Various means of swapping weapon types is within the scope of the present invention. For example, a variety of “skins” could be implemented, such as shells that attach or slide over the projectile-firing device **104**. These skins could have the necessary electronics to tell the blaster what weapon type to become. The skins could combine both physical and virtual objects in accordance with a particular weapon type.

(118) Other objects could be utilized to provide users **116**, **118** with varying attributes for in-game performance. For example, stickers containing passive NFCs can be provided as packs, and players can affix them to objects of their choice to place around the battlefield, allowing loot drops including weapon types and upgrades; these objects could be misplaced without much loss, as new stickers can be easily replaced. In addition, scopes or silencers may be provided as accessories to the device **104** to provide relative effects on gameplay. For example, the scope could increase the range at which a particular weapon type is effective.

(119) FIG. 7 is a flow diagram of an extended-reality projectile-firing gaming method **200**. At Step **202**, a gaming session in accordance with a predetermined gameplay configuration is initiated. The extended-reality gaming application is configured, in a preferred embodiment, to control and communicate instructions and/or messages to initiate the gaming session. Alternatively, or in combination, the remote device **112** is configured to initiate the gaming session.

(120) At Step **204**, projectile-firing device data is received from one or more blasters (e.g., projectile-firing devices **104**, **152**.) The projectile-firing device data preferably includes data configured to update gaming metric data in accordance with a gameplay configuration. The gameplay configuration may specify requirements for the gaming session, including available health points, shields, blaster types, available rounds, reload interval times, rounds per magazine, and battlefield objectives, among other things. The gaming metric data in a preferred embodiment tracks player scores, updating the gaming metric data to confirm successful impacts (e.g., projectile hits) from particular blasters **104** against particular wearable **106** and non-wearable battlefield objects **108**, **148**, **172**, **176**, etc. Thus, the projectile-firing device data includes data indicating that

a projectile was fired, the time the projectile was fired, and location and orientation information. The projectile-firing device data may also include information obtainable from one or more sensors, such as near-field communication, radio frequency, infrared, magnetometer, and global positioning satellite sensory data.

(121) At Step **206**, battlefield object data is received from one or more battlefield objects such as wearable or non-wearable devices. The battlefield object data preferably includes data configured to update gaming metric data in accordance with a gameplay configuration. Thus, the battlefield object data preferably includes information that the battlefield object **106, 108, 148, 172, 176**, etc., was impacted with a fired projectile, the time of impact, and the location and orientation of the battlefield object. The battlefield object data may also include information obtainable from one or more sensors, such as near-field communication, radio frequency, infrared, magnetometer, and global positioning satellite sensory data. The battlefield object data may wholly, or partly, be provided via an affixed remote device **112** configured to provide the sensory data.

(122) As described above, at Steps **204** and **206**, the projectile-firing device data and the battlefield object data are received. In a preferred embodiment, these data are received by the cloud-based computing system **110** via the network **102** and usable by the extended-reality gaming application. In other embodiments, these data are received by a standalone application at the physical location of the gaming session, or may be a combination of cloud-based and physical location-based computing and processing. One or more remote computing centers or systems may be utilized to receive and process the data.

(123) At Step **208**, the firing of a projectile is confirmed, and a time stamp is associated with the firing of the projectile. For example, the data tracks the projectiles as they are fired by the projectile-firing device **104**, and provides time stamps when each projectile was fired in real- or near-real-time. At Step **210**, the location and orientation of the blaster that fired the projectile are determined. As described above, the sensory data from the projectile-firing device data is preferably utilized to determine the location and orientation of the blaster.

(124) At Step **212**, an impact on a battlefield object is confirmed, and a time stamp is associated with the impact on the battlefield object. For example, the battlefield object is tracked to perceive impacts on wearable devices **106** and/or non-wearable devices **108, 148, 172, 176**, etc. At Step **214**, the location and orientation of the battlefield object are determined. As described above, the sensory data from the projectile-firing device data is preferably utilized to determine the location and orientation of the battlefield object.

(125) At Step **216**, it is determined whether the time the projectile was fired and the time of the impact on the battlefield object are within a predetermined time period. For example, the predetermined time is measured in milliseconds, as the blaster's projectile velocity is preferably at least 90 FPS and can exceed 200 FPS. This determination provides confirmation that the blaster shot the particular projectile that was detected by the particular battlefield object.

(126) At Step **218**, it is determined whether the location of the blaster that fired the projectile and the battlefield object detecting the impact are at locations within a predetermined area. As described herein, the physical range of the blaster **104** is limited, but the blaster **104** may be further limited by the application such that certain impacts outside of the predetermined area are outside of the permitted blaster's **104** range. The location determination permits further confirmation that the blaster shot the particular projectile that was detected by the particular battlefield object.

(127) At Step **220**, it is determined whether the orientation of the blaster that fired the projectile is oriented with the orientation of the battlefield object detecting an impact by a projectile. If the orientation is within a predetermined direction or orientation, the orientation determination permits further confirmation that the blaster shot the particular projectile that was detected by the particular battlefield object.

(128) At Step **222**, the gaming metric data is updated confirming a successful impact on the battlefield object and a successful hit by the blaster **104**. The updated gaming metric data attributes

a decrease in health points to a user or battlefield object that was impacted by the blaster **104**, in accordance with the gaming configuration, and attributes damage inflicted upon the user or battlefield object to the user associated with the blaster **104**. The gaming metric data in a preferred embodiment tracks player scores, updating the gaming metric data to confirm successful impacts (e.g., projectile hits) from particular blasters **104** against particular wearable **106** and non-wearable battlefield objects **108, 148, 172, 176**, etc. in real- or near-real-time.

(129) The determinations in Steps **216, 218**, and **220** and the updating of the gaming metric data in Step **222** are preferably conducted using the gaming application.

(130) FIG. **8** is a flow diagram of an extended-reality projectile-firing gaming method **300**. At Step **302**, blaster data is received from a blaster **104**. The blaster data is preferably the projectile-firing device data associated with method **200**. At Step **304**, wearable vest data is received from a wearable vest **106**. The wearable vest data is preferably the battlefield object data associated with method **200**. At Step **306**, battlefield object data is received from a battlefield object **108, 148, 172, 176**, etc. The battlefield object data is preferably the same battlefield object data associated with method **200**.

(131) At Step **308**, the time of firing of a fired projectile from the blaster **104** is recorded. Preferably the time of firing is recorded and stored in connection with the gaming application. At Step **310**, an impact of the fired projectile on the wearable vest **106** is confirmed. The methodology for determining an impact on a wearable vest **106** is disclosed elsewhere herein. At Step **312**, the time of impact of the fired projectile on the wearable vest **106** is recorded. At Step **314**, an impact of the fired projectile on the battlefield object **108, 148, 172, 176**, etc. is confirmed. The methodology for determining an impact on a battlefield object **108, 148, 172, 176**, etc. is disclosed elsewhere herein. At Step **316**, the time of impact of the fired projectile on the battlefield object **108, 148, 172, 176**, etc. is recorded.

(132) At Step **318**, it is determined whether the time of impact on the wearable vest **106** and the time of firing the fired projectile is within a predetermined time period. This determination provides confirmation that the blaster shot the particular projectile that was detected by the particular wearable vest. At Step **320**, it is determined whether the time of impact on the battlefield object **108, 148, 172, 176**, etc. and the time of firing the fired projectile is within a predetermined time period. This determination provides confirmation that the blaster shot the particular projectile that was detected by the particular battlefield object. For example, the predetermined time is measured in milliseconds, as the blaster's projectile velocity is preferably at least 90 FPS and can exceed 200 FPS.

(133) At Step **322**, a successfully fired projectile attributed to the blaster **104** is recorded. Preferably the application records the successful hit of the projectile and associates the successful hit with the blaster **104** and the associated user. At Step **324**, a successful impact on the wearable vest **106** and/or the battlefield object **108, 148, 172, 176**, etc. is recorded. Preferably the application records the successful impact of the projectile on either the wearable vest **106** and/or the battlefield object **108, 148, 172, 176**, etc. In practice, a blaster's **104** projectile will likely only be attributable to a single wearable vest **106** or a single battlefield object **108, 148, 172, 176**, etc. However, when another projectile-firing device such as a Bazooka, mine, or grenade involves splash damage, one or more wearable vests **106** and/or battlefield object **108, 148, 172, 176**, etc. may be impacted by a single "explosion."

(134) At Step **326**, the gaming metric data is updated to record the successful impact and the successful hit and attribute each to corresponding devices and their associated users.

(135) It will be appreciated by one of ordinary skill in the art that method **300** does not specifically include location and/or orientation confirmation of successful impacts or hits.

(136) However, those confirmations may be implemented in the method **300**. Alternatively, any combination of timing, location, and orientation data and/or information may be provided to confirm successful impacts or hits, and are within the scope of the present invention.

(137) It will also be appreciated that the updated gaming metric data identified in the methods **200**, **300** preferably is provided to the projectile-firing devices **104**, **152**, the display **154**, and/or the HUD display **142**, and/or the like, to provide updated gaming score information in accordance with the gaming configuration during the gaming session in real- or near-real-time.

(138) Referring now to FIG. 9, FIG. 9 is a smart target **400** in accordance with a preferred embodiment of the present invention. The smart target **400** is configured to be integrated and utilized with a smart target co-witnessing hit attribution system in accordance with a preferred embodiment of the present invention. In an embodiment, the smart target **400** is a standalone target **108** or a monolithic target **148**. The smart target **400** preferably includes a piezoelectric (PE) sensor **410** configured to determine an impact of a particular projectile upon (or near) the smart target. In an embodiment, the smart target **400** is arranged as a “flower.” The standalone target **400** includes a pistil portion **402** (e.g., “pistil” of flower) and a plurality of petal portions **404** (e.g., “petals” of flower). As shown in FIG. 9, the smart target **400** includes six petal portions **404**, but in other embodiments, the number of petal portions **404** may be greater or fewer than six. Each of the pistil portion **402** and the petal portions **404** are configured to separately determine whether a projectile has impacted that portion of the standalone target **400**.

(139) In an embodiment, a PE sensor **410** is embedded or integrated and/or operably coupled with each of the pistil portion **402** and the petal portions **404**. In another embodiment, a single PE sensor **410** or two or more PE sensors **410** are utilized to sense impacts on the pistil portion **402** and the petal portions **404**. A trained model, as described further herein, is preferably utilized to enable electronic control circuitry to determine when a projectile has impacted a particular portion of the smart target **400**. Depending on the application it may be advantageous to provide multiple PE sensors **410** for a single smart target **400**. In an embodiment, the same target impact methodology is utilized for the wearable device targets **136** and the central and outer portions **138**, **140** of the standalone target **108**. The smart target **400** includes support members **406** to support the pistil portion **402** and the petal portions **404**. The smart target **400** also includes a stand **408** configured to support the smart target **400** in an upright position. One of ordinary skill in the art would understand that a PE sensor **410** is an electromechanical device that uses the piezoelectric effect to measure changes in pressure, strain, force (or other physiological effects) by converting them to an electrical charge or signal. In smart target parlance, the PE sensor **410** is utilized for hit detection. In other embodiments, sensors other than the PE sensor **410** may be implemented, as described herein.

(140) As an example, the smart target **400** is affixed to one or more petals of a target system. In an embodiment, the smart target depicted in FIG. 9 includes a central PE sensor **410** and outer “petal” PE sensors **410** configured to detect hit impacts. The same (or similar) modular PE sensor **410** is also configured to be used in connection with wearable vests, for example. Referring now to FIG. 10, FIG. 10 is a perspective view of a wearable device **412** including one or more smart targets **400** in accordance with a preferred embodiment of the present invention. One can see that the PE sensor **410** that forms the center (e.g., central PE sensor **410**) and petals (outer “petal” PE sensors **410**) of the wearable device **412** are affixed in the central torso portion of the wearable vest depicted in FIG. 2. In addition, the wearable device **412** may include one or more PE sensors **410** on the back of the user's wearable vest or on shoulder portions of the wearable device. In an embodiment, the wearable device **412** is a smart target **400**. Depending on the size and sensitivity of the PE sensors, as well as electronic, firmware, or software filtering of the PE sensor's electrical charge (i.e., electrical signal), one or more PE sensors **410** may be trained to determine a successful impact of a projectile (e.g., gellet).

(141) In a preferred embodiment, logic is trained using machine learning (ML) to identify positive strikes against background noise and artifact impacts by identifying the waveform signature (i.e., electrical signal) specific to gellet composition and anticipated material density. For example, logic trained through machine learning obtains training data; a robot (or the like) uses a smart blaster **416**

and varies distance, capturing data to understand the waveform signature that a gellet (or other projectile) generates upon a successful hit. In an embodiment, a two-dimensional convolutional ML model may be trained by multiple sensor inputs being provided simultaneously. The captured data provides a model to be utilized by the smart target **400** to detect a successful hit. The hit detection trained model accounts for off-angle impacts, variance of impact pressure, and other physical metrics (i.e., relative location adjacent to the smart target **400**) to increase or decrease the amount of “damage” points given to the user upon a successful hit. The machine learning element obtains data for the trained model used by the PE sensor to understand the angle of impact and range of a gellet impact, as but one example.

(142) More specifically, the trained model is generated using machine-learning-assisted noise reduction (e.g., ambient vibration, non-projectile impacts), which may be implemented in hardware, firmware, software, or a combination thereof. Machine learning also may include identification of different materials that may be advantageous to eliminate or reduce noise reduction and provide a “cleaner” waveform signature to be used in creating filtering or the like. Envelope detection and peak detection circuits may be provided for determining characteristic gellet impact signals. This feature is not limited to gellet projectiles, but can be used for all types of projectiles including airsoft, paintball, bow style projectiles, firearms, etc.

(143) The trained model also reduces the ability for “false positive” impacts or cheating, distinguishing non-truthed impacts from real impacts. The accuracy embodied by the trained model eliminates or reduces the necessity for a referee to determine which individual was hit by a projectile, and attribute the hit to another individual. In an embodiment, the smart target **400** includes a filter and amplifier circuit tuned to impact responses for detecting vibration. For example, the filter and amplifier circuit may be configured to eliminate portions of a signal from the PE sensor **410** that are not attributable to a hit by a projectile. As described above, the trained model permits the designer of the filter and amplifier circuitry to understand, based on many multiple projectiles hitting a target with a PE sensor **410**, the waveform signature of a successful hit. Thus, the filter and amplifier circuit is preferably designed to eliminate background noise and non-projectile impacts, able to differentiate between real impacts by a projectile and everything else that may occur during gameplay.

(144) In a preferred embodiment, the smart target **400** includes an IR detection sensor **414**. Hit detection may be implemented such that if the PE sensor **410** identifies a hit, but the IR detection sensor **414** does not, then the detected hit is not counted as a real strike. In another embodiment, the PE sensor **410** (i.e., hit detection) versus IR sensor **414** (i.e., hit recognition) may be expressed as a ratio and a successful hit is not counted unless the ratio exceeds a particular threshold.

(145) In an embodiment, the IR detection sensor **414** may also confirm the angle at which a projectile hit the target, which in a gameplay environment, may adjust “damage” or “hit points” up or down depending upon the angle of impact, tracked and calculated by the application.

(146) In an embodiment, The IR sensor **414** can be embedded with PE sensor **410**. In other embodiments, the IR sensor **414** could be in a different location from the PE sensor **410**.

Depending on the application; i.e., wearable device **412** vs standalone smart target **410**, it may be advantageous to locate the IR sensor **414** near the PE sensor **410**, or distanced apart. For example, the distance between the IR sensor **414** and the PE sensor **410** preferably is from ½ centimeter to 2 meters. In addition, there may be advantages to the location of the IR sensor **414** versus the PE sensor **410** at varying ranges. For example, logic trained with machine learning may indicate that optimum distances between the sensors is achievable at ranges in accordance with the application. For example, in a blaster-based gaming environment, it may be advantageous to train logic at distances most likely to be experienced during gameplay (e.g., 50 feet or less).

(147) Referring still to the IR sensor **414**, any combination of carrier frequency, emitter LED, and receiver module can be used. IR messaging modules are preferably universal and included to interface with other IR devices. IR modules preferably consist of a high power infrared LED meant



for line of sight transmission systems, and an infrared receiver module with matched optical characteristics. The IR modules combined with application features include IR-based universal interaction with the application for actions such as shoot to interact, shoot to pair.

(148) Other sensors or methodologies may be utilized alternatively or in combination with the PE sensor **410** and IR sensor **414** co-witnessing techniques. For example, a brain-computer interface (BCI) based impact detection system may include skin contact electroencephalogram (EEG) sensors (or others) read players' physiological response to being hit by a projectile count it as a hit in the system. The BCI technique may be utilized in combination with IR cowitnessing or other confirmation methods. In addition, computer-vision (CV) aided hit detection may be utilized. For example, cameras or other RF sensitive equipment may watch for projectile impact and use machine learning (ML) to filter noise and false positives. The CV-aided hit detection methodology may be best suited to controlled arena play where external cameras/sensors can be permanently mounted, and cost concerns are mitigated. The CV-aided hit detection methodology utilizes video and/or photo recognition technology to help accuracy and gameplay experience.

(149) Referring still to FIG. **10**, a BCI sensor **458** is depicted encircling the head of the user **116**, **118**. the BCI sensor **458** includes EEG sensors **460** configured to record an electrogram of spontaneous electrical activity of the brain. While the EEG sensors **460** are depicted in certain areas of the BCI sensor **458**, they are placed for illustrative purposes only, and one of ordinary skill in the art would recognize that the EEG sensors **460** would be optimally placed in order to accurately monitor and record electrical activity of the brain. For example, a MyndBand EEG headset may be configured to be used in connection with the system disclosed herein. Other headsets or devices are within the scope of the present invention. In other embodiments, a full or partial “helmet” may be utilized to facilitate placement of the EEG sensors **460** on portions of the head of the user **116**, **118** that are not covered by the “headband” style headset depicted in FIG. **10**.

(150) As with the PE sensor **410**, the BCI sensor **458** preferably is configured to filter or reduce noise of the EEG sensors **460** signals by utilizing a trained model. Similar to the descriptions of ML-trained models herein, a blaster **104** is shot multiple times from varying distances and angles to a smart target **400** and data is gathered from the BCI sensor **458** in the form of EEG sensors **460** signals. A trained model can assist the control circuitry of the smart target **400** (or distributed/remote software application and/or circuitry) to filter out false positives by isolating portions or bands of frequencies. This will assist the system in quickly and efficiently determining whether an object that hits the user is a projectile, and may further allow the system to determine impact angle and distance. For example, the signal from the EEG sensors **460** may appear different (e.g., different distribution of frequencies, or reduced intensity at certain frequency bands) at an impact angle of 0° versus 45°. Likewise, the signal from the EEG sensors **460** may appear different at close ranges versus farther ranges (e.g., different intensity or reduced intensity at greater distances).

(151) In a study using a MyndBand EEG headset, EEG monitoring as a means of hit tracking and co-witnessing was researched. The MyndBand, as well as other EEG systems, uses thin and flexible electrodes (e.g., EEG sensors **460**) in direct skin contact with areas of the user's head. The flexible electrodes may be mounted in hats, headbands, Gel Blaster masks, or other wearables. The EEG sensors **460** were connected to a small computing device with a Bluetooth transmitter. The resulting data (e.g., signals from the EEG sensors **460**) were read and displayed in real-time on a computer or mobile device. The EEG sensors **460** produced signals that was parsed to indicate brainwave changes associated with being hit by a gellet (e.g., projectile).

(152) The study utilized a surprise impact methodology such that the participants were not anticipating being hit by a gellet. The participants were instructed to proceed with routine tasks until a tester used a blaster **104** to hit the users with a gellet. The study results showed a significant change in the EEG sensors **460** signal using different participants upon being hit by a gellet.

(153) In a second phase of the study, the participants were again struck by gellets by surprise, but

only half of the time. The other half of the time, the participants would experience the tester shooting the gellet near the participant, but not actually hit the participant. The data from the EEG sensors **460** signals reveals that additional significant brain activity occurred only when the gellet actually made contact.

(154) In addition, the participants were test shot both to the front of the body and the back of the body to ensure the element of surprise. The EEG sensors **460** signals were recorded in real-time. The results indicated that almost all shots captured at short range over a significant amount of time (e.g., 1-2 minutes) successfully showed significant brain activity.

(155) At varying ranges and thus, varying intensities, the EEG sensors **460** signals may vary in their intensity. It may be advantageous to utilize the BCI sensor **458** data as a primary or secondary co-witnessing element (e.g., along with the PE sensor **410**, the IR sensor **414**, or other suitable co-witnessing elements) depending on the range and reliability of the EEG sensors **460** signals at those ranges. The control circuitry or remote computing device can prioritize the BCI sensor **458** data depending on the gameplay environment, ranges, and other factors.

(156) Still further, a fabric deformation hit detection methodology may be utilized to provide co-witnessing of an impact. "Smart" textile wearables may detect changes in electrical current when deformed from projectile impact. For example, optical fibers or conductive fabrics embedded or weaved into a wearable device **412** or other wearable shirt or vest are within the scope of the present invention. As with the above methodologies, the fabric deformation hit detection methodology may be an additional form of co-witnessing a hit, and may also be utilized with ML for noise (false positive) reduction. In other embodiments, the fabric deformation hit detection methodology may be implemented through use of electronic textiles in which microcontrollers, sensors, and actuators may be integrated into wearable items.

(157) In an embodiment, when one player is closer than a threshold close-quarters distance, the application can turn off gellet (projectile) firing and utilize IR only for hit attribution.

(158) FIG. **11** is a smart blaster **416** in accordance with a preferred embodiment of the present invention. The smart blaster **416** includes the mechanical and electronic structures described herein to shoot gellets and project IR radiation. The smart blaster **416** preferably includes a wireless networking device configured to transmit and receive data as a node in a wireless network (e.g., mesh network, as described herein). In other embodiments,

(159) Hit attribution, in a preferred embodiment, is a combination of hit detection and hit recognition. For example, hit detection includes the ability for a physical object with a PE sensor **410** to detect when the physical object was hit, as described above. In an embodiment, hit detection determines whether an object that strikes at or near the PE sensor **410** is a projectile. Hit detection may also include timer logic between smart blaster **416** and target to associate time of impact to distance of shooter. In addition, triangulation utilizing a mesh network (as more fully described herein) improves higher accuracy in hit detection. These features are used either independently or in combination to verify a correct strike by a correct individual and/or smart blaster **416** associated with the individual.

(160) Hit recognition, in a preferred embodiment, enables co-witnessing of hit detection such that even greater accuracy in hit attribution may be achieved. For example, the smart target **400** includes embedded infrared (IR) detection sensors **414** that may detect hits from multiple angles. The smart blaster **416** (or other projectile-firing device) includes the ability to generate and project (e.g., emit) IR radiation in a beam **420** from an IR emitter **418**. The IR emitter **418** includes a beam angle at which the IR radiation is projected that provides a spread/angle over distance. For example, the range of an IR emitter **418** is 50 feet and the optics allow a spread of about 2 degrees, which preferably provides a beam spread of 2 feet at a distance of 50 ft. In another embodiment, the IR emitter **418** spread may be defined as a wider diameter meant to illuminate at an angle where the top of the beam cone corresponds to a straight trajectory, and the bottom of the beam cone corresponds to a smart blaster **416** pointed towards the sky at a certain angle that makes the

projectiles have a longer range, and thus the IR beam **420** accounts for projectile drop. A combination of these techniques could be used to accommodate different game modes and play styles. In an embodiment, the smart blaster **416** may include a modifiable emitter **418** that includes two beams **420** having different spread angles. A point blank shot may be a wider beam spread (e.g., 3 degrees) with a larger spread angle while a longer range shout may be a narrower beam spread (e.g., 2 degrees) with a smaller spread angle. The emitter **418** may be configured to automatically switch between the different spread angles

(161) IR-based hit detection sensors **414** embedded in the smart target **400**, when coupled with smart blasters **416**, can maintain bidirectional communication across multiple players and use timer-based co-witnessing to identify, arbitrate, and attribute hits to the corresponding shooter. The smart target technology may be implemented on smart blasters **416** (e.g., emitters **418**) and all other objects, including wearables and standalone targets (e.g., receivers **414**).

(162) FIG. **12** is a schematic diagram of a circuit for the smart target co-witnessing hit attribution system in accordance with a preferred embodiment of the present invention. The circuit of FIG. **12** is configured to be utilized in connection with the petal portion **404**, the outer petals of the smart target **400** depicted in FIG. **9**. The circuit depicted in FIG. **11** includes a microcontroller **422** (e.g., ATTINY404SSMCSOIC), a PE sensor **410**, a programming chip (e.g., UPDI programming chip) **424**, interfaces **426** (e.g., SD1 and SM04B-SRSS-TB(LF)(SN) interfaces), an array of addressable LEDs **428** (e.g., WS2812s), and an IR sensor **414**.

(163) The microcontroller **422** depicted in FIG. **11** is a Microchip ATtiny404SSMC-type microcontroller. One of ordinary skill in the art would understand that other microcontrollers are within the scope of the present invention without departing from the present invention. The ATtiny404SSMC microcontroller utilizes high-performance, low-power AVR® RISC architecture. The microcontroller **422** is capable of running up to 20 MHz, with up to 2/4 KB Flash, 128/256 bytes of SRAM, and 64/128 bytes of EEPROM in a 14-pin package. The microcontroller **422** utilizes a flexible and low-power architecture including Event System and SleepWalking, accurate analog features, and advanced peripherals. The microcontroller **422** includes PIEZO\_IN and IR\_SENSOR for piezo sensor **410** and IR sensor **414** inputs.

(164) The PE sensor **410** is a piezoelectric sensor configured to detect impacts at or near the physical location of the PE sensor **410**. The programming chip **424** is a Microchip Unified Program and Debug Interface (UPDI) configured for external programming and debugging of the device. The interfaces **426** are SD1 and SM04B-SRSS-TB(LF)(SN) interfaces that are configured to receive I2C SCL and SDA signals. The SCL and SDA signals are data and clock signals, respectively. The array of addressable LEDs **428** is configured to drive LEDs or LED strips to light up portions of the petal portions **404**. One of ordinary skill in the art would understand that other components may be utilized to achieve the electronic circuit and functionality represented by the circuit of FIG. **12**, including a different programming chip **424**, interfaces **426**, and addressable LEDs **428**.

(165) FIGS. **13-1**, **13-2**, and **13-3** depict portions of a schematic diagram of a circuit for a smart target co-witnessing hit attribution system in accordance with a preferred embodiment of the present invention. The circuit of FIGS. **13-1**, **13-2**, **13-3** is configured to be utilized in connection with a “pistil,” the center portion **402** of the smart target **400** of FIG. **1**. The circuit of FIGS. **13-1**, **13-2**, and **13-3** include a microcontroller **430**, a switch **432**, a USB driver **434**, a fuse **436**, a USB receptacle **438**, an external battery **440**, an alternate microcontroller **442**, an IR sensor **414**, a PE sensor **410**, interfaces **444**, and an array of addressable LEDs **446**. Instead of an ATTINY404SSMCSOIC microcontroller **422**, an ESP32-WROOM-32E microcontroller **430** is shown in FIG. **13-1**. The ESP32-WROOM-32E microcontroller **430** is a Wi-Fi+Bluetooth+Bluetooth LE MCU module that is configured for low-power sensor networks and voice encoding, music streaming and MP3 decoding. The ESP32-WROOM-32E includes a PCB antenna. One of ordinary skill in the art would understand that other microcontrollers are within the

scope of the present invention without departing from the present invention. The microcontroller **430** includes PIEZO\_IN and IR\_SENSOR for the PE sensor **410** and the IR sensor **414** inputs. FIG. **13-1** also depicts the switch **432**, configured to switch between the microcontroller **430** and the alternate microcontroller **442**, the USB driver **434** configured to provide power to the circuit, a fuse **436** in connection with power operation, the USB receptacle **438** configured to receive a USB power input, and an external battery **440** configured to power the device when not connected to USB power. Designations of A-G indicate connections of the circuit to the components shown on FIGS. **13-2** and **13-3**.

(166) FIG. **13-2** shows an alternate microcontroller **442** (e.g., ESP32-C3-WROOM-02-N4) in place of the previously described microcontroller **430**. The ESP32-C3-WROOM-02 microcontroller **442** includes Wi-Fi and Bluetooth LE modules, 4 MB external SPI flash, and an on-board PCB antenna. FIG. **13-2** also depicts the PE sensor **410** and the IR sensor **414** as well as the array of interfaces **444** (as opposed to single SD1 and SM04B-SRSS-TB(LF)(SN)) interfaces).

(167) The portion of the circuit depicted in FIG. **13-2** also includes a filtering resistor **445** to filter the PE sensor **410** signal before it is input into the microcontroller **442**. In an embodiment, the filtering resistor **445** to filter the PE sensor **410** signal in FIG. **13-2** is chosen so that the peak of the detected signal output is set to a certain voltage. If the value of this resistor is changed, the peak voltage shown by the projectile impact will change as well. Filtering to increase detection accuracy may involve including a circuit that allows changing this resistor value in real time to decrease or increase the device sensitivity. In another embodiment, the IR emitter **418** spread may be defined as a wider diameter meant to illuminate at an angle where the top of the beam cone corresponds to a straight trajectory, and the bottom of the beam cone corresponds to a smart blaster **416** pointed towards the sky at a certain angle that makes the projectiles have a longer range, and thus the IR beam **420** accounts for projectile drop. A combination of these techniques could be used to accommodate different game modes and play styles.

(168) Other filtering techniques could be implemented in the circuit or in firmware/software. For example, an envelope detector to wrap a gellet waveform may be utilized. A peak detector circuit may also be utilized, in which an output only would be provided to the microcontroller as a high signal when the voltage is above a peak voltage threshold. The output of a “high” signal may also reduce sampling requirements. A bandpass filter may also be utilized to tune the circuit and isolate the waveform signature. Combinations of these filtering techniques could be utilized. The ML trained logic is utilized to determine the best filtration techniques to fit the particular application (e.g., gellet impact on a standalone target).

(169) Hit attribution technologies embedded in the smart target and associated circuitry, firmware, and software can be expanded to include other input sources such as smartphone microphones (or standalone microphones). For example, identification and inventory of gellet acoustic signatures may be differentiated against background noise and artifacts for increased accuracy. The acoustic signatures may be added to the PE sensor **410**/IR sensor **414** logic to achieve increased accuracy, or any combination of these hit attribution sources of data may be used separately to provide an appropriate level of accuracy, depending on a variety of factors including complexity of the device, cost, manufacturing difficulty, or other factors. A smartphone could also provide other sources of data to the application, including GPS or other sensor data to integrate to achieve increased accuracy of hit attribution. In place of a smartphone, or in combination, a wearable or standalone device could implement the other sensor data to increase accuracy. For example, a standalone device at a known location relative to the impact location may be utilized to determine distance (e.g., echos), to provide additional sensor data for the application to utilize to increase accuracy.

(170) FIG. **13-3** shows the array of addressable LEDs **446** (e.g., WS2812s) and transistors **447** coupled to the data control input/output of the USB driver **434**.

(171) In a preferred embodiment, to provide real-time or near-real-time score tracking, it is advantageous to categorize the trained model from ML datasets as a key from a database and

efficiently broadcast the key as a hexadecimal identifier over low-latency communications. This “minimal” approach achieves efficient transmission and reception of data to be utilized by the application to track scores and provide scoring data to players in a gaming environment.

(172) FIGS. **14-17** depict exemplary smart target **400** mesh networks in accordance with preferred embodiments of the present invention. The target mesh networks depicted in FIGS. **14-17** may be utilized independently or in combination with each other. The target mesh networks of FIGS. **14A** and **14B** are built around a Bluetooth Low Energy (BLE) topology/protocol. The BLE protocol may utilize mesh networking to implement many-to-many communication. FIG. **14A** depicts a basic BLE mesh network including a router **448**, phone **450**, root **452**, and a plurality of nodes **454**. The BLE mesh network in connection with an embodiment of the present invention is configured to control multiple Bluetooth-enabled devices from a single root **452**. In the network topology represented in FIG. **14A**, the root **452** controls all three nodes **454** by use of a publish/subscription model. The child nodes **454** are subscribed to the parent node **454** and the parent node **454** is publishing messages to the child nodes **454**.

(173) FIG. **14B** also depicts a basic BLE mesh network with a router **448**, a phone **450**, a root **452**, nodes **454**, and relay nodes **456**. The relay nodes **456** are utilized to propagate child nodes **454** and move data downstream.

(174) The target mesh networks of FIGS. **15A**, **15B**, **15C** are built around an ESP-NOW topology/protocol. The ESP-NOW protocol enables multiple devices to communicate with one another without using Wi-Fi, and is similar to low-power 2.4 GHz wireless connectivity utilized in certain low-power devices. In FIG. **15A**, there is shown a Broadcast Mode in which small packet messages are broadcast to all devices from the root **452** to the nodes **454**. The message packets include sender ID and payload. In FIG. **15B**, there is shown a Direct Message mode in which a direct message is sent from a root **452** to a device (e.g., node **454**). In FIG. **15C**, there is shown a Game State Broadcast mode in which a game state broadcast is sent from each device (node **454**) in accordance with a predetermined interval. The current game state payload contains target hit counts and target miss counts. When receiving a Game State Broadcast message, the smart target **400** sets its active state after comparing the incoming message with its current state. The higher counts are then set as the target's active state. The target then sends a Game State Broadcast.

(175) The target mesh networks of FIGS. **16A**, **16B**, **16C** consist of a self-organizing and self-healing WiFi mesh network. FIG. **16A** illustrates a self-organizing mesh network in which a node **454** is configured to connect with other nodes autonomously. In an example, each of the nodes **454** depicted in FIG. **16A** are communicating with each other within the mesh network. The mesh network depicted in FIG. **16A** permits multiple devices (i.e., nodes **454**) to communicate with each other without the need for a router **448** through a wireless local area network. FIG. **16B** illustrates a node **454** that is out of range (shown as a shaded node **454**), and thus, not connected to the mesh network. When the node **454** comes back into range, communication with the node **454** is reestablished because of the constant provisioning protocol (as more fully described herein). FIG. **16C** shows a connection to a node **454** (shown as a shaded node **454**) that was out of range and lost connection, but reestablishes communication with a node **454** when it is within range of that particular node **454**. Once communication is reestablished, the formerly out of range node **454** is able to transmit and receive data with the mesh network. This methodology is referred to as “self-healing” and can be maintained within a self-organizing mesh network autonomously.

(176) The target mesh networks of FIGS. **17A**, **17B**, **17C** are WiFi mesh networks. FIG. **17A** illustrates a basic WiFi mesh network in which the effective range of a router **448** is extended. Multiple nodes **454** within the network permit data to be transferred across the entire network, even if a particular node **454** may have been out of range of the router **448**. The line depicted in the graphic illustrates the range of the router **448**, in which only the root **452** is within range; however, the nodes **454** are configured to be connected to the root **452** (and other nodes **454**) within the network to maintain connectivity and the exchange of data. FIG. **17B** illustrates a mesh network

where the router **448** and router **452** are integrated, thereby eliminating the necessity for a separate router **448**. FIG. **17C** illustrates a mesh network with a backup router **456**, configured such that a node **454** of the network is also configured to be a router **448**. Thus, if the first router **448** in the network loses functionality, the backup router/node **456** may be utilized to continue to facilitate communication with the nodes **454**.

(177) FIGS. **14-17** illustrate preferred embodiments of the present invention. Preferably, the smart target **400** system will adapt a wireless connection module-type that allows for multiple smart targets **400** to be connected to the same network and function as nodes **454** to selectively increase gameplay complexity, as well as expand the network coverage area, thus increasing the gameplay area size. Mesh networking will also uniquely allow for the low-latency broadcast of messages and commands to all players **116**, **118** and devices in the network despite line-of-sight limitations existent in other systems.

(178) In preferred embodiments, achieving real-time or near-real-time communication via a mesh network may include WiFi without traditional TCP/IP protocol, simplified message IDs, and/or device IDs. For example, simple messaging transmits single packets blindly, or until an acknowledgement is received on its complementary receiver. Data link messaging may establish a connection through simple messaging, then begins to transmit continuously through an error-corrected protocol.

(179) In a preferred embodiment, low-latency communications may be implemented through a unique protocol. For example, the smart target system (and blasters) make use of a unique protocol to transmit information throughout the network with minimal data package size. The reduction of bandwidth required and usage of high-throughput wireless networks can broadcast information that pertains to gameplay in a number of different factors and controls.

(180) FIG. **18** is a table of a message protocol **458** in accordance with a preferred embodiment of the present invention. The message protocol **458** depicted in FIG. **18** includes “Dev Mode” **460** and “Game Mode” **462** for smart targets **400** and the application **464**. “Dev Mode” **460** is a developer's mode **460** that includes Commands **466** for the smart targets **400** such as Set Target Index **468**, Clear All Hit Count **470**, Time Between Cycles **472**, Cycles Between Hits **474**, Start Slice Min Value **476**, Minimum Read Value **478**, and Hit Read Value **480**. Commands for the application **464** include Sensor Hit Count **482**, Dev App Settings **484**, and Dev Sensor Readings **486**. The “Game Mode” **462** is a mode used to facilitate communication between nodes **454** during a gameplay session that includes Commands **466** for the smart targets **400** such as Reset All Targets **488**, Activate Target **490**, and Color Test **492**. Commands **466** for the application **464** include Hit Message **494**, Miss Message **496**, and Active State **498**. The Dev Mode **460** is for developers or programmers of the gameplay environment and development of features for gameplay. The Dev Mode **460** is also utilized to develop future physical functions of the smart targets and smart target systems. The Dev Mode **460** is preferably configured to develop ML trained logic for use with the PE sensor **410** or other sensors as described herein.

(181) The Game Mode **462**, as described herein, includes separate Target **400** and App **464** protocols **458**. Referring to the Game Mode **462** for the App **464**, in a preferred embodiment, once a smart target **400** is hit, the target **400** will broadcast a message with an ID **500**/Parameters **502** structure (i.e., the Hit Message Command **494**). For example, the hexadecimal ID **500** for a Hit Message Command **494** is 0xAA, identifying the message type. The Parameters **502** portion of the protocol **458** is an array of hexadecimal values (indices) configured to keep track of a hit count for each of the seven targets' hit counts **504** in the gameplay mode (e.g., Target 1 Hit Count . . . Target 7 Hit Count; Target 1 Miss Count . . . Target 7 Miss Count). The Player ID **506** portion of the Parameters **502** portion of the protocol **458** identifies a player associated with the target hit. The application/client **464** is configured to recognize each ID **500** type of the protocol **458** and configured to receive and recognize a particular number of Parameters **502** and process them accordingly. For example, the Hit Message **494** displays the number of times each designated panel

has been hit by gelllets on the mobile app. The Miss Message **496** displays the number of times gelllets were fired but did not hit the designated (and connected) panel on the mobile app. The Target Miss Count **508** Command **466** is needed for scoring in individual game modes as some games will take into account the accuracy of a user's **116**, **118** fired gelllets. The Active State **498** shows which of the panels are activated or not. The App **464** Game Mode **462** therefore is configured to receive and display the Hit Message **494**, Miss Message **406**, and Active State **498** in accordance with embodiments of the present invention as described herein.

(182) Referring still to FIG. **18**, the Target **400** Game Mode **462** includes Reset All Targets **488**, Activate Target **490**, and Color Test **492** Commands **466**. The Reset All Targets **488** Command **466** resets each of the connected panels to the null or a deactivated state. The Activate Target **490** Command **466** activates all of the connected panels to be ready for the game to begin. The Color Test **492** Command **466** lights up the panels in a connected smart target with a custom color from individual Red, Blue, and Green hex values. In a preferred embodiment, the custom colors are determined in accordance with the Target States of FIG. **19**.

(183) The Activate Target **490** Command **466** includes Target State **510** Parameters **502** Target 0 State **510**, Target 1 State **510** . . . Target 6 State **510**. FIG. **19** is a table of smart target **400** Target States **510** in accordance with a preferred embodiment of the present invention. In accordance with FIG. **19**, there is disclosed **256** possible Target States **510** each specified by an ID **512** in a hex value. For example, the “Off” **516** Target State **510** is associated with the ID **512** 0x00 hex value. Each ID **512** and Target State **510** are also associated with a Color **514**; for example, the ID **512** of 0x00 and Target State **510** “Off” **516** is associated with Color **514** #000000 (i.e., no color). FIG. **19** identifies other Target States **510** (i.e., State column) of Hit **518**, Miss **520**, BT (i.e., Bluetooth) not connected **522**, BT connected **524**, Player 1 **527**, and Player 2 **528**. Other Target States **510** are possible in accordance with embodiments of the present invention. The Color **514** column of the Target States **510** protocol identifies the hex value (e.g., hex code) of the color, and in parentheses, the actual color associated with the hex value. For example, Magenta **530** is ascribed a hex value of #FF00FF, Green **532** is ascribed a hex value of #00F00, Red **534** is ascribed a hex value of #FF0000, Yellow **536** is ascribed a hex value of #FFFF00, Cyan **538** is ascribed a hex value of #00FFFF, and Blue **540** is ascribed a hex value of #0000FF. For example, each hex value indicates amounts of Red/Green/Blue. The hex value #FF00FF specifies an RGB value of R=255, G=0, and B=255. In an embodiment, the Color Test **492** Command **466** is a way to test specific colors by providing the red, green, and blue hex values. The Target States **510** in FIG. **19** have a predefined color assignment and only give information about the current state of each panel. Target State **510** will tell the app **464** whether the panel has received a miss or a hit, it will also communicate the Bluetooth connection state of the smart target **400** device as a whole, among other states.

(184) The Parameters **502** portion of the App **464** Game Mode **462** for the Activate Target **488** Command **466** includes Target 1 State **510** . . . Target 7 State **510**, and Player ID **506**. The Target 1 State **510** . . . Target 7 State **510** identifies the current Target State **510** in accordance with FIG. **19**. The Player ID **506** is the identifier for each connected smart blaster **416**.

(185) Still referring to FIG. **19**, there is illustrated the target states **510** of a smart target **400** in accordance with a preferred embodiment of the present invention. In an embodiment, the ‘Off’ **516** state represents the panel is turned off and not detecting impacts. The ‘Hit’ **518** state is communicated when the panel is hit by a gelllet. The ‘Miss’ **520** state is communicated when the panel is missed by a gelllet (by time out or by hitting another panel). The ‘BT not connected’ state **522** is communicated when the smart target **400** is not connected to a user's phone with Bluetooth. The ‘BT connected’ state **524** is communicated when the smart target **400** is connected to a user's phone with Bluetooth. The ‘Player 1’ state **526** is communicated when the smart target **400** is connected to a user's phone and the user is denoted as Player 1 in the mobile app. The ‘Player 2’ state **528** is communicated when the smart target is connected to a user's phone and the user is denoted as Player 2 in the mobile app.

(186) FIG. 20 is a table of a message protocol 542 in accordance with a preferred embodiment of the present invention. FIG. 20 depicts a message protocol 542 that is an alternative embodiment to the message protocol 458 of FIG. 18. In FIG. 20, the “Game Mode” 462 is a mode used to facilitate communication between nodes 454 during a gameplay session that includes Commands 466 for the smart targets 400 such as Reset All Targets 488, Activate Target 490, similar to those same commands depicted and described in connection with FIG. 18. Commands 466 for the application 464 include Hit Message 494, Miss Message 496, and Active State 498. The Commands 466, ID 500, and Parameters 502 portions of the message protocol 542 generally function similarly to the same portions of the message protocol 458, as further described herein. Notably, however, the Parameters 502 portion of the message protocol 542 use the nomenclature “Pad” instead of “Target.” However, the nomenclature “Pad” can be substituted for the term “Target” as the “Pad” is a portion or area of the smart target 400 such as the pistil portion 402 and the petal portions 404.

(187) In addition, the message protocol 542 includes additional Commands 466 relating to the Target 400 portion of the protocol 542 such as Deactivate Target 544, Activate Pad 546, Activate Hidden Pad 548, Set All Color 550, Set Color State 552, Set Device Name 554, and Set Device Mode 556. The Deactivate Target 544 Command 466 is configured to deactivate the smart target 400, conceivably in a scenario in which, according to the gameplay configuration, the target has been taken out by gellet hits and is no longer active. The Activate Pad 546 Command 466 is configured to activate a portion or area of the smart target 400, such as only the pistil portion 402 or one of the petal portions 402. The Activate Hidden Pad 548 Command 466 is configured to activate a pad (e.g. pistil or petal portions 402, 404) that has previously been hidden in accordance with the gameplay configuration by the software application. The Set All Color 550 Command 466 is configured to set the color of each of the pads to a specified RGB hex value. The Set Color State 552 is configured to set the color of specified pads (e.g., Pad 0 State) to a particular color. The Set Device Name 554 Command 466 is configured to specify a particularized name for a device (e.g., node 454 or smart target 400) using an array of hex values. The Set Device Mode 556 Command 466 is configured to set a Device Target Mode 560 in accordance with the Target Modes 560 depicted in the table of FIG. 21B.

(188) The message protocol 542 includes the additional Command 466 relating to the App 464 portion of the protocol 542 of Display State 558. The Display State 558 Command 466 is configured to display the particular Target State 510. In an embodiment, the Target State 510 is displayed on the application 464 to the user 116, 118.

(189) FIG. 21A is a table of smart target 400 Target States 510 in accordance with a preferred embodiment of the present invention. In accordance with FIG. 21A, there is disclosed 256 possible Target States 510 each specified by an ID 512 in a hex value. For example, the “Off” 516 Target State 510 is associated with the ID 512 0x00 hex value. Each ID 512 and Target State 510 are also associated with a Color 514; for example, the ID 512 of 0x00 and Target State 510 “Off” 516 is associated with Color 514 #000000 (i.e., no color). FIG. 21A identifies other Target States 510 (i.e., State column) of Hit 518, Miss 520, BT (i.e., Bluetooth) not connected 522, BT connected 524, Player 1 527, and Player 2 528. Other Target States 510 are disclosed as “(Active)” states. The Color 514 column of the Target States 510 protocol identifies the hex value (e.g., hex code) of the color, and in parentheses, the actual color associated with the hex value. Each of the colors depicted in FIG. 19 are repeated in FIG. 21A. In addition, Orange 562 is ascribed a hex value of #FF4000, Light Green 564 is ascribed a hex value of #C9FF00, and Teal 566 is ascribed a hex value of #00FF2A.

(190) FIG. 21B is a table of smart target 400 Target Modes 560 in accordance with a preferred embodiment of the present invention. The Target Modes 560 are configured to operate the colors of the smart targets 400 in accordance with a prescribed animation or gameplay configuration. For example, an ID 568 and State 570 indicate the specific hex value and associated information about the state of the Target Mode 560. A Rainbow Animation 572 is configured to display a rainbow



display of color on the smart target **400**. A Flipper Game **574**, a Eliminator Game **576**, and a IR Dev Game **578** are all configured to operate colors and timing of the colors in connection with a particular gameplay configuration. One of ordinary skill in the art would understand that a variety of animations and gameplay configurations are within the scope of the present invention and those depicted in FIG. **21B** are exemplary.

(191) FIG. **22A**, **22B** is a flow diagram of a smart target co-witnessing hit attribution method **600** in accordance with a preferred embodiment of the present invention. At Step **602**, a first sensor signal from a first sensor is received in response to an impact at or near a first smart target location. For example, the first sensor is the PE sensor **410**. In other embodiments, the first sensor is an IR sensor **414**, an EEG sensor **460** (or BCI sensor **458**), a computer-vision sensor, a radio-frequency sensor, or a smart textile sensor. While the first sensor is specified to be a single sensor, other embodiments in which multiple PE sensors **410** or IR sensors **414** or combinations of the above-referenced sensors are provided. The first sensor signal is triggered to be transmitted to and received by control circuitry preferably integrated with the first sensor upon sensing an impact. As disclosed herein, a PE sensor **410** generates an electrical charge or signal that corresponds to the physical act that it sensed, and provides the signal for processing. The impact may be the impact of any object, including any type of projectile. In an embodiment, it is determined whether the impact occurs at or near a predetermined distance (e.g., an area) from a predetermined location (e.g., the first smart target location). The predetermined distance may be set such that impacts only on the first smart target location (e.g., a pistil portion **402** or petal portion **404**) are provided to the control circuitry. In another embodiment, the predetermined distance may encompass a larger area than the first smart target location. In a case in which the first sensor senses an impact on a second smart target location, that data may be utilized to confirm impact on the second smart target location.

(192) The control circuitry may be configured to determine that the first sensor signal corresponds to a successful impact. For example, the PE sensor **410** may be configured through circuitry (or otherwise) to provide only signals beyond a peak threshold to the control circuitry. In an embodiment, the control circuitry is integrated with a peak detector and threshold. Alternatively, the first sensor may simply provide all signals in response to an impact at or near the first smart target location. In this embodiment, other methods such as software or firmware manipulation or filtration may be provided to assure that the first sensor signal is an appropriate detection of a hit in association with the first smart target location. Still further, filtration techniques as described herein may be utilized to isolate and assist the control circuitry in determining that the impact was a projectile associated with the gameplay configuration and environment rather than another type of impact. For example, the projectile is a gellet.

(193) The first smart target location preferably is a pistil portion **402** or petal portion **404**. The PE sensor **410** alone, or in combination with control circuitry, may be configured to sense more than one smart target **400** location. For example, a single first sensor may be utilized to sense impacts of projectiles at two locations on the smart target **400**. The first smart target location may also be a smart target **400** located on the wearable device **412**.

(194) At Step **604**, a second sensor signal from a second sensor is received in response to an impact at or near the first smart target location. Step **604** is configured to co-witness the same impact event as that identified in connection with Step **602**. As with Step **602**, the second sensor may be any of a variety of sensors including a PE sensor **410**, an IR sensor **414**, an EEG sensor **460** (or BCI sensor **458**), a computer-vision sensor, a radio-frequency sensor, or a smart textile sensor. Preferably, if the first sensor is a PE sensor **410** then the second sensor is an IR sensor **414**, or another type of sensor as described above. However, in an embodiment the first and second sensors are both PE sensors **410**.

(195) As with Step **602**, at Step **604**, the second sensor signal is generated based in response to an impact at or near the first smart target location. If the second sensor signal is an IR sensor **414** signal, and the first sensor signal is a PE sensor **410** signal, then the IR sensor **414** may be

configured to provide an IR signal should it receive radiation at a particular wavelength. For example, the wavelength is 940 nanometers. Other wavelengths are suitable for use as an IR sensor **414**. The IR sensor **414** in an embodiment does not utilize infrared radiation, but rather, utilizes frequencies and wavelengths outside of the infrared band. One of ordinary skill in the art would understand that while certain wavelengths are theoretically possible, one should provide safe solutions that prevent dangerous situations for a user's eyes and other body parts sensitive to certain types of radiation.

(196) At Step **606**, the first sensor signal and the second sensor signal were determined to be received within a predetermined time period. In an embodiment, the first and second sensor signals are presumed to arrive almost instantly from their respective sensors. One of ordinary skill in the art would understand how to provide the first and second sensor signals to control circuitry, or the like, to ensure that the time between sensing the impact and the time the signals are received is the same or substantially the same between the sensors. The predetermined time period preferably is set to be sufficiently small (e.g., fractions of seconds/milliseconds) to ensure that the signals were generated at or about the same time. This is important to ensure that the first and second sensors co-witnessed the same impact at the same time. For example, if the first sensor and the second sensor are the same (e.g., two IR sensors **414**), then the signals would typically be generated at around the same time. However, if the first sensor is the PE sensor **410** and the second sensor is the IR sensor **414**, then the predetermined time period would necessarily be longer due to the time the projectile takes to impact a smart target **400**. Preferably, a predetermined time period depends upon several factors including circuitry delay, the relative delays caused by the circuitry for different types of sensors, the time the projectile takes to impact a target, and the velocity of the device **416**, all of which are within the scope of the present invention.

(197) At Step **608**, a smart target update is provided indicating a successful hit in connection with the first smart target location of the smart target **400**. For example, the smart target update is configured to identify the target state of each smart target location of the smart target. Preferably, utilization of the target mesh networks and protocols described herein ensure sufficiently fast communication to ensure real-time or near-real-time updates to the smart target states.

(198) At Step **610**, a third sensor signal from a third sensor is received in response to an impact at or near a second smart target location. For example, the third sensor is the PE sensor **410**. In other embodiments, the third sensor is an IR sensor **414**, an EEG sensor **460** (or BCI sensor **458**), a computer-vision sensor, a radio-frequency sensor, or a smart textile sensor. While the third sensor is specified to be a single sensor, other embodiments in which multiple PE sensors **410** or IR sensors **414** or combinations of the above-referenced sensors are provided. The third sensor signal is triggered to be transmitted to and received by control circuitry preferably integrated with the third sensor upon sensing an impact. As disclosed herein, a PE sensor **410** generates an electrical charge or signal that corresponds to the physical act that it sensed, and provides the signal for processing. The impact may be the impact of any object, including any type of projectile.

(199) The control circuitry may be configured to determine that the third sensor signal corresponds to a successful impact of a projectile. For example, the PE sensor **410** may be configured through circuitry (or otherwise) to provide only signals beyond a peak threshold to the control circuitry. In an embodiment, the control circuitry is integrated with a peak detector and threshold. Alternatively, the third sensor may simply provide all signals in response to an impact at or near the second smart target location. In this embodiment, other methods such as software or firmware manipulation or filtration may be provided to assure that the third sensor signal is an appropriate detection of a hit in association with the second smart target location. Still further, filtration techniques as described herein may be utilized to isolate and assist the control circuitry in determining that the impact was a projectile associated with the gameplay configuration and environment rather than another type of impact. For example, the projectile is a gellet.

(200) The second smart target location preferably is a pistil portion **402** or petal portion **404**. The

PE sensor **410** alone, or in combination with control circuitry, may be configured to sense more than one smart target **400** location. For example, a single third sensor may be utilized to sense impacts of projectiles at two locations on the smart target **400**. The second smart target location may also be a smart target **400** located on the wearable device **412**.

(201) At Step **612**, a fourth sensor signal from a fourth sensor is received in response to an impact at or near the second smart target location. Step **612** is configured to co-witness the same impact event as that identified in connection with Step **610**. As with Step **610**, the second sensor may be any of a variety of sensors including a PE sensor **410**, an IR sensor **414**, an EEG sensor **460** (or BCI sensor **458**), a computer-vision sensor, a radio-frequency sensor, or a smart textile sensor. Preferably, if the third sensor is a PE sensor **410** then the fourth sensor is an IR sensor **414**, or another type of sensor as described above. However, in an embodiment the third and fourth sensors are both PE sensors **410**.

(202) As with Step **610**, at Step **612**, the fourth sensor signal is generated based in response to an impact at or near the second smart target location. If the fourth sensor signal is an IR sensor **414** signal, and the third sensor signal is a PE sensor **410** signal, then the IR sensor **414** may be configured to provide an IR signal should it receive radiation at a particular wavelength. For example, the wavelength is 940 nanometers. Other wavelengths are suitable for use as an IR sensor **414**. The IR sensor **414** in an embodiment does not utilize infrared radiation, but rather, utilizes frequencies and wavelengths outside of the infrared band. One of ordinary skill in the art would understand that while certain wavelengths are theoretically possible, one should provide safe solutions that prevent dangerous situations for a user's eyes and other body parts sensitive to certain types of radiation.

(203) The second smart target location is preferably distinct from the first smart target location, but located on the same smart target **400**. However, in other embodiments, the first and second smart target locations are located on different smart targets **400**. The first through fourth sensors and control circuitry are preferably configured such that a single impact at or near both the first and second smart target locations will generate signals in which either the first and second sensor signals confirm an impact or the third and fourth sensors confirm an impact, but not both. Stated otherwise, the sensors should be sophisticated enough so that impacts of projectiles on first and second smart target locations may be distinguished.

(204) At Step **614**, the third sensor signal and the fourth sensor signal were determined to be received within a predetermined time period. In an embodiment, the third and fourth sensor signals are presumed to arrive almost instantly from their respective third and fourth sensors. One of ordinary skill in the art would understand how to provide the third and fourth sensor signals to control circuitry, or the like, to ensure that the time between sensing the impact and the time the signals are received is the same between the sensors. The predetermined time period preferably is sufficiently small (e.g., fractions of seconds/milliseconds) to ensure that the signals were generated at or about the same time. This is important to ensure that the third and fourth sensors co-witnessed the same impact at the same time. For example, if the third sensor and the fourth sensor are the same (e.g., two IR sensors **414**), then the signals would typically be generated at around the same time. However, if the third sensor is the PE sensor **410** and the fourth sensor is the IR sensor **414**, then the predetermined time period would necessarily be longer due to the time the projectile takes to impact a smart target **400**. Preferably, a predetermined time period depends upon several factors including circuitry delay, the relative delays caused by the circuitry for different types of sensors, the time the projectile takes to impact a target, and the velocity of the device **416**, all of which are within the scope of the present invention.

(205) At Step **616**, a smart target update is provided indicating a successful hit in connection with the second smart target location of the smart target **400**. For example, the smart target update is configured to identify the target state of each smart target location of the smart target. Preferably, utilization of the target mesh networks and protocols described herein ensure sufficiently fast

communication to ensure real-time or near-real-time updates to the smart target states.

(206) At Step **618**, multiple smart target updates are received in real-time or near-real-time. In an embodiment, the smart target updates are provided in Steps **608** and **616** by control circuitry in connection with a wireless network in accordance with one or more protocols. For example, the target mesh networks disclosed herein and the protocols identified as communicating via hexadecimal values may be utilized to provide or transmit the smart target updates. Preferably, the smart target updates are target state updates in accordance with the protocols disclosed in FIGS. **18-21B**. The smart target updates are configured to be received by an extended-reality gaming application. The extended-reality gaming application is, for example, the application **464**.

(207) FIG. **22B** is a second portion of a flow diagram of a smart target co-witnessing hit attribution method **600** in accordance with a preferred embodiment of the present invention. FIG. **22B** shows the continuation of FIG. **22A**, indicating “A” from Step **218**. At Step **620**, projectile-firing device data is received, the data including orientation and location of the device **416**, and the time of firing of the device **416**. In an embodiment, the device **416** includes a smart target **400**. The device **416** preferably includes an IR emitter **418** that emits IR radiation in a beam **420**. The IR emitter **418** module or other components in the device **416** may supply the orientation and location of the device **416**.

(208) At Step **622**, the time of projectile impact and the time of firing the device **416** is determined to be within a predetermined time period. Depending on the speed of the network, it is possible to determine at Step **622** whether the particular smart target update and the projectile-firing device data were received within a predetermined time period, particularly where a target mesh network is utilized in connection with the protocols of FIGS. **18-22B**. In other embodiments, a time stamp may be provided along with the first through fourth sensor signals, or a time stamp may be generated by control circuitry and packetized along with the smart target updates. Still further, in other embodiments, a sensor and network latency may be calculated and predetermined such that receipt of the smart target update and the projectile-firing device data include a predetermined time delay from which it is determined whether the time of projectile impact and the time of firing occurred within the predetermined time period. The predetermined time period preferably is sufficiently small (e.g., fractions of seconds/milliseconds) to ensure that the co-witnessing of the impact and the attribution of the hit to the particular device **416** should be correlated. It may be advantageous to determine the predetermined time period based on a relatively immediate time window between emission of IR radiation from the IR emitter **418** and detection by the IR sensor **414**. In other embodiments, it may also be advantageous to calculate an expected time a projectile will travel over the distance from the device **416** to the smart target **400** and determine the predetermined time period based on the time window between firing of the projectile and the detection of an impact of an object by the PE sensor **410**. For example, with a known distance to the target (e.g., 25 feet) and a known velocity (e.g., 170 FPS), the time to target for the projectile is approximately 147 milliseconds. The predetermined period of time therefore can be set to 147 milliseconds or greater to provide a margin of error (e.g., 150 ms or 160 ms). A timer can be set to expire at the predetermined period of time and a successful hit attributed both to the device **416** and the portion of the smart target **400** hit by the projectile if the time of impact is within the predetermined period of time. One of ordinary skill in the art would understand that a variety of methodologies to determine the predetermined time period are within the scope of the present invention.

(209) A variety of methods are within the scope of the present invention to determine shooter identity. In an embodiment, impact detection by the PE sensor **410** is turned off, or in “sleep mode,” and the portion of the smart target **400** in question is set to only actively detect IR radiation. When the IR radiation is received, a timer is started on the processor associated with the smart target **400** circuitry and the impact detection is activated. The timer is permitted to run for a predetermined period of time, usually less than a few seconds, with the most optimal time

determined by the physical distance between the device **416** and the PE sensor **410**. If the timer runs over the limit set in the code, the impact detection is turned off and the IR message received can be counted as a “miss.” If the PE sensor **410** detects an impact of a projectile, then the IR message is correlated with the PE sensor **410** detection and counted as a successful “hit.”

(210) In another embodiment, to time the IR message reception with the impact detection, one or more separate sensors are provided to accurately measure distance in real-time or near-real-time. For example, a LIDAR sensor or laser-based sensor capable of measuring distances of up to 50 feet (e.g., a device **416** effective range) is provided. This distance measurement can be used by itself, or in combination with an integrated photogate-based chronograph integrated into the device **416**. In a preferred embodiment, the barrel **129** is molded and/or integrated in an infrared wavelength-invisible material with two pairs of an infrared LED pointed across the barrel facing a phototransistor sensitive to that wavelength. These two pairs of sensors will trigger once a projectile passes through them, and a timer is started when the first photogate sensor is triggered, and stopped when the second photogate sensor is triggered. This permits individual projectile muzzle velocity measurement, which can calculate the trajectory characteristics, such as horizontal velocity, which together with the distance measurement, can produce an estimation for the time that the Gellet is in flight. This estimation can be used to fine tune the IR detection vs impact detection time window (e.g. predetermined time period).

(211) In another embodiment, in order to identify individual shooters aiming at the same target, a variety of these techniques can be applied. In one embodiment, a hard-coded time threshold can serve as the limiting factor in identifying which shooter achieved the first impact, where two shooters positioned at different distances from the target will have their projectiles arrive at different times, with one arriving after the target's threshold expires, thus not being counted.

(212) In yet another embodiment, if both (or more) shooters are aiming from the same distance at the same target, the target mesh network can relay notices of multiple IR messages (e.g., IR sensor **414** detections) from different devices **416** being received at the same location on the smart target **400** back to the same devices **416** and any other “referee” devices in the target mesh network. A central processor handling hit attribution would then digitally stagger the projectile fire from these devices **416**, to give the timing algorithm additional gaps between shots and IR messages across the devices **416**, thus increasing shooter identification accuracy.

(213) In another preferred embodiment, the devices **416** can be outfitted with location-tracking sensors such as ultra-wideband (UWB) or Bluetooth Low Energy (BLE) that can selectively turn on or off the IR sensors physically positioned around the smart target **400** or wearable device **412**, correlating the localization sensor data with the IR sensor directional data, further increasing accuracy in detecting the correct shooter.

(214) At Step **624**, the device **416** is determined to be within a predetermined range of the first smart target location. Given that the projectile-firing device data includes the location of the device **416**, a calculation of distance between the device **416** and the first smart target location can be made. The location may be fixed in an arena environment or may be dynamic where the location is on a wearable device **412**. The target mesh network may be utilized to provide triangulation between different nodes **454** so that a location may be determined relative to other nodes within the network. Alternative methodologies of location services may be utilized, such as GPS tracking or the like. The predetermined range may be preset based on the battlefield environment and gaming characteristics. For example, the predetermined range may be 50 feet, 40 feet, 25 feet, 10 feet, or other ranges depending upon the range of the device **416** and the distances of the smart targets **400**. The predetermined range may also be determined based on a calculation of time between a time the device **416** is shot and a time the smart target **400** (or a location on the smart target **400**) is impacted by a projectile. Using a known device **416** velocity (e.g., 90 FPS), a calculation of the distance to the target may be determined. Once the distance is known, a predetermined range may be the distance plus an additional amount of range (e.g., as a margin of error). Other methodologies

to determine distance (and therefore, predetermined range) are disclosed herein, such as using a LIDAR sensor. One of ordinary skill in the art would understand that a variety of methodologies to predetermine the range are within the scope of the present invention.

(215) At Step **626**, the device **416** is determined to be oriented towards the first smart target location. Using the orientation information from the projectile-firing device data, and the relative location of the first smart target location, the application **464** determines if a vector based on the orientation of the device **416** is aligned with a predetermined area of the first smart target location. For example, IR detection of the device **416** IR emitter **418** may occur at the first smart target location if the smart target **400** includes an IR sensor **414**. In that case, the IR emitter **418** emits radiation in a beam **420** that spreads over distance, as more fully described herein. Thus, if the beam **420** is detected by the IR sensor **414** of the smart target **400** at the first smart target location, then a match between the particular device **416** and the first smart target location of the smart target **400** is achieved. The predetermined area may be 2 feet, as the typical range of an IR emitter **418** is 50 feet and the optics allow a spread of about 2 degrees, which preferably provides a beam spread of 2 feet at a distance of 50 ft. The predetermined area may also be larger or smaller depending on the beam spread of the IR emitter **418**. For example, a wider beam spread would typically mean a larger predetermined area while a narrower beam spread would typically mean a smaller predetermined area.

(216) At Step **628**, gaming metric data is updated to indicate a successful hit by the device **416** on the first smart target location of the smart target **400**. The updated gaming metric data preferably is configured to be updated and usable by the application **464**. Steps **630**, **632**, and **634** are substantively similar to Steps **624**, **626**, and **628**, except that instead of the first smart target location a second smart target location is an object of the determinations. It should be understood that the same device **416** may be utilized to hit both the first smart target location and the second smart target location. However, preferably the same projectile may not be determined to hit both the first and second smart target locations simultaneously. If different projectiles hit the first and second smart target locations, respectively, then those impacts will have occurred at slightly different times, thereby matching the correct projectile impact with the correct smart target location. One of skill in the art would understand that only Step **622** may be utilized to match projectile impacts with a particular smart target location. However, when Step **622** is utilized also with Steps **624**, **626**, **628** or Steps **630**, **632**, **634**, then higher accuracy may be accomplished. One of ordinary skill in the art would also understand that a variety of other sensors (e.g., three or more) may be provided to achieve higher accuracy. It is also the case that certain sensors may be better suited for particular ranges, gaming configurations, or devices. For example, it may be advantageous to use the BCI sensor **458** only at certain ranges to ensure that sufficient impacts occur on the user's **116**, **118** body. In addition, at low-impact gaming configurations in which the barrel tip **131** is in a reduced velocity mode, it may be advantageous to utilize sensor types other than the BCI sensor **458**. In addition, it may be advantageous to co-witness a first impact with an IR sensor **414**, since certain gaming modes may automatically turn off the ability to fire a projectile at a distance closer than 5 feet from any users **116**, **118**, for example.

(217) At Step **636**, a player state is updated to attribute the hit to a user associated with the device **416**. The application **464** is configured to update the gaming metric data and display the data to a user (and on particular smart targets **400**). For example, the hit may increment a hit count by 1 unit; in other embodiments, as described herein, the hit may count for a particular amount of hit points. The application **464** is configured to take the angle of impact and range into account, as well as the gaming configuration and “ammo” type used, to calculate and attribute a certain amount of hit points to the user.

(218) At Step **638**, a smart target state is updated to attribute the hit against an appropriate target location of the smart target **400**. The application **464** is configured to update the gaming metric data and display the data to a user (and on particular smart targets **400**). For example, the hit may

increment a hit count against the first and/or second smart target locations associated with a smart target **400**. As above with Step **636**, in Step **638**, application **464** is configured to take the angle of impact and range into account, as well as the gaming configuration and “ammo” type used, to calculate and attribute a certain amount of hit points against a particular location on a target. As described herein, the locations may be pistil portions **402**, petal portions **404**, or standalone smart targets **400** affixed to the wearable device **412**. Other embodiments with smart targets **400** affixed or located on other battlefield objects are within the scope of the present invention.

(219) In addition to the foregoing, an IR sensor **414** is configured to provide an IR signal that represents a particular angle of detection and an intensity of the infrared radiation. Based on the angle of detection and the intensity of the infrared radiation, the method **600** may further determine whether the device **416** is within a predetermined margin of error with respect to the location of the first smart target location and the orientation of the device **416**. Stated otherwise, the IR sensor **414** may provide the pertinent information correlating the first smart target location and the device **416** to increase the accuracy of the hit attribution between a user and a smart target **400**. In addition, it may be possible to eliminate a need to obtain orientation and location information from the device **416** each time it is fired, so long as the angle of detection and intensity information is sufficient to gauge the orientation and location of the device **416** relative to the first smart target location.

(220) In a preferred embodiment of the present invention, functionality is implemented as software executing on a server that is in connection, via a network, with other portions of the system, including databases and external services. The server comprises a computer device capable of receiving input commands, processing data, and outputting the results for the user. Preferably, the server consists of RAM (memory), hard disk, network, central processing unit (CPU). It will be understood and appreciated by those of skill in the art that the server could be replaced with, or augmented by, any number of other computer device types or processing units, including but not limited to a desktop computer, laptop computer, mobile or tablet device, or the like. Similarly, the hard disk could be replaced with any number of computer storage devices, including flash drives, removable media storage devices (CDs, DVDs, etc.), or the like.

(221) The network can consist of any network type, including but not limited to a local area network (LAN), wide area network (WAN), and/or the internet. The server can consist of any computing device or combination thereof, including but not limited to the computing devices described herein, such as a desktop computer, laptop computer, mobile or tablet device, as well as storage devices that may be connected to the network, such as hard drives, flash drives, removable media storage devices, or the like.

(222) The storage devices (e.g., hard disk, another server, a NAS, or other devices known to persons of ordinary skill in the art), are intended to be nonvolatile, computer readable storage media to provide storage of computer-executable instructions, data structures, program modules, and other data for the mobile app, which are executed by CPU/processor (or the corresponding processor of such other components). The various components of the present invention, are stored or recorded on a hard disk or other like storage devices described above, which may be accessed and utilized by a web browser, mobile app, the server (over the network), or any of the peripheral devices described herein. One or more of the modules or steps of the present invention also may be stored or recorded on the server, and transmitted over the network, to be accessed and utilized by a web browser, a mobile app, or any other computing device that may be connected to one or more of the web browser, mobile app, the network, and/or the server.

(223) References to a “database” or to “database table” are intended to encompass any system for storing data and any data structures therein, including relational database management systems and any tables therein, non-relational database management systems, document-oriented databases, NoSQL databases, or any other system for storing data.

(224) Software and web or internet implementations of the present invention could be accomplished with standard programming techniques with logic to accomplish the various steps of

the present invention described herein. It should also be noted that the terms “component,” “module,” or “step,” as may be used herein, are intended to encompass implementations using one or more lines of software code, macro instructions, hardware implementations, and/or equipment for receiving manual inputs, as will be well understood and appreciated by those of ordinary skill in the art. Such software code, modules, or elements may be implemented with any programming or scripting language such as C, C++, C#, Java, Cobol, assembler, PERL, Python, PHP, or the like, or macros using Excel or other similar or related applications with various algorithms being implemented with any combination of data structures, objects, processes, routines or other programming elements.

(225) Although the operations of the method(s) herein are shown and described in a particular order, the order of the operations of each method may be altered so that certain operations may be performed in an inverse order or so that certain operations may be performed, at least in part, concurrently with other operations. In another embodiment, instructions or sub-operations of distinct operations may be implemented in an intermittent and/or alternating manner.

(226) Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” As used herein, the terms “connected,” “coupled,” or any variant thereof, means any connection or coupling, either direct or indirect, between two or more elements; the coupling of connection between the elements can be physical, logical, or a combination thereof. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description of the Preferred Embodiments using the singular or plural number may also include the plural or singular number respectively. The word “or” in reference to a list of two or more items, covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list.

(227) The above-detailed description of embodiments of the disclosure is not intended to be exhaustive or to limit the teachings to the precise form disclosed above. While specific embodiments of and examples for the disclosure are described above for illustrative purposes, various equivalent modifications are possible within the scope of the disclosure, as those skilled in the relevant art will recognize. Further, any specific numbers noted herein are only examples: alternative implementations may employ differing values, measurements or ranges.

(228) Although the operations of any method(s) disclosed or described herein either explicitly or implicitly are shown and described in a particular order, the order of the operations of each method may be altered so that certain operations may be performed in an inverse order or so that certain operations may be performed, at least in part, concurrently with other operations. In another embodiment, instructions or sub-operations of distinct operations may be implemented in an intermittent and/or alternating manner.

(229) The teachings of the disclosure provided herein can be applied to other systems, not necessarily the system described above. The elements and acts of the various embodiments described above can be combined to provide further embodiments. Any measurements or dimensions described or used herein are merely exemplary and not a limitation on the present invention. Other measurements or dimensions are within the scope of the invention.

(230) Any patents and applications and other references noted above, including any that may be listed in accompanying filing papers, are incorporated herein by reference in their entirety. Aspects of the disclosure can be modified, if necessary, to employ the systems, functions, and concepts of the various references described above to provide yet further embodiments of the disclosure.

(231) These and other changes can be made to the disclosure in light of the above Detailed Description of the Preferred Embodiments. While the above description describes certain



embodiments of the disclosure, and describes the best mode contemplated, no matter how detailed the above appears in text, the teachings can be practiced in many ways. Details of the system may vary considerably in its implementation details, while still being encompassed by the subject matter disclosed herein. As noted above, particular terminology used when describing certain features or aspects of the disclosure should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features or aspects of the disclosure with which that terminology is associated. In general, the terms used in the following claims should not be construed to limit the disclosures to the specific embodiments disclosed in the specification unless the above Detailed Description of the Preferred Embodiments section explicitly defines such terms. Accordingly, the actual scope of the disclosure encompasses not only the disclosed embodiments, but also all equivalent ways of practicing or implementing the disclosure under the claims.

(232) While certain aspects of the disclosure are presented below in certain claim forms, the inventors contemplate the various aspects of the disclosure in any number of claim forms. For example, while only one aspect of the disclosure is recited as a means-plus-function claim under 35 U.S.C. § 112, ¶6, other aspects may likewise be embodied as a means-plus-function claim, or in other forms, such as being embodied in a computer-readable medium. (Any claims intended to be treated under 35 U.S.C. § 112, ¶6 will include the words “means for”). Accordingly, the applicant reserves the right to add additional claims after filing the application to pursue such additional claim forms for other aspects of the disclosure.

(233) Accordingly, although exemplary embodiments of the invention have been shown and described, it is to be understood that all the terms used herein are descriptive rather than limiting, and that many changes, modifications, and substitutions may be made by one having ordinary skill in the art without departing from the spirit and scope of the invention.

## Claims

1. An extended-reality projectile-firing gaming system comprising: a projectile-firing device, a battlefield object, a network configured to provide communication and control connectivity in accordance with at least one protocol to the projectile-firing device and the battlefield object, an extended-reality gaming application configured to control the projectile-firing device and the battlefield object in accordance with a gaming session having a gameplay configuration for providing an extended-reality environment, receive projectile-firing device data from the projectile-firing device, the projectile-firing device data indicating that a projectile was fired, receive battlefield object data from the battlefield object, the battlefield object data indicating that an impact occurred on the battlefield object, and an extended-reality device comprising at least one display and configured to provide computer-aided vision of real and virtual objects and a computer-aided interface for interaction with the real and virtual objects, wherein the extended-reality gaming application is further configured to update gaming metric data to indicate a successful impact of the fired projectile on the battlefield object and a successful hit by the projectile-firing device on the battlefield object when a time the impact occurred on the battlefield object and a time the projectile was fired occur within a predetermined time period, wherein the predetermined time period is selected to confirm that the projectile-firing device fired the fired projectile that was detected as impacting the battlefield object, and wherein the updated gaming metric data is configured to modify the real and virtual objects in real or near real-time, and receive a location and orientation of the projectile-firing device, wherein the extended-reality gaming application records a successful impact of the fired projectile on the battlefield object and a successfully fired projectile by the projectile-firing device associated with the battlefield object when the location of the projectile-firing device is within a predetermined range from the battlefield object and the orientation of the projectile-firing device is aligned at a predetermined area associated with the battlefield object.

2. The extended-reality projectile-firing gaming system of claim 1 wherein the computer-aided interface comprises voice-activation control configured to select one or more of the real and virtual objects and to provide the commands or instructions.
3. The extended-reality projectile-firing gaming system of claim 1 wherein the computer-aided interface comprises an eye sensor configured to monitor eye contact with the real and virtual objects.
4. The extended-reality projectile-firing gaming system of claim 3 wherein the commands or instructions are selectable using a control input when the eye contact is aligned with one of the real and virtual objects.
5. The extended-reality projectile-firing gaming system of claim 4 wherein the control input is voice-activated.
6. The extended-reality projectile-firing gaming system of claim 4 wherein the control input is at least one sensor configured to monitor hand movements and the commands or instructions are selectable based on the monitored hand movements.
7. The extended-reality projectile-firing gaming system of claim 4 wherein the control input is a remote device.
8. The extended-reality projectile-firing gaming system of claim 7 wherein the remote device is coupled to the projectile-firing device.
9. The extended-reality projectile-firing gaming system of claim 7 wherein the remote device is configured to receive data via the network from at least one of the projectile-firing device and the battlefield object.
10. The extended-reality projectile-firing gaming system of claim 4 wherein the control input is a remote device configured to be affixed to the projectile-firing device and provide at least one of near-field communication, radio frequency, infrared, magnetometer, and global positioning satellite sensory data to the extended-reality gaming application.
11. The extended-reality projectile-firing gaming system of claim 3 wherein the eye sensor identifies a selection of the commands or instructions associated with the real and virtual objects when the eye contact is fixed on one of the real and virtual objects for at least a first predetermined time period.
12. The extended-reality projectile-firing gaming system of claim 3 wherein the eye sensor identifies a selection of the commands or instructions associated with the real and virtual objects when the eye contact is fixed on one of the real and virtual objects for at least a first predetermined time period but does not exceed a second predetermined time period.
13. The extended-reality projectile-firing gaming system of claim 1 wherein at least one of the real and virtual objects is displayed in response to the projectile-firing device data or the battlefield object data.
14. The extended-reality projectile-firing gaming system of claim 1 wherein at least one of the real and virtual objects is displayed in response to the battlefield object data.
15. The extended-reality projectile-firing gaming system of claim 1 wherein the gameplay configuration comprises information corresponding to at least one of a game mode, a health meter, a shield meter, a projectile tally, a projectile repository tally, a rate of fire, a reload interval, a virtual power up, a success tally, a failure tally, a success to failure ratio, an alert, a battlefield role, a team affiliation, a battlefield objective, and a set of battlefield parameters.
16. The extended-reality projectile-firing gaming system of claim 1 wherein a type of the projectile-firing device is selectable through the computer-aided interface.
17. The extended-reality projectile-firing gaming system of claim 1 wherein the commands and instructions comprise one or more of an amount of projectiles available for firing, a predetermined velocity of the projectiles for firing, a predetermined rate of fire of the projectiles, and a predetermined reload interval of the amount of projectiles.
18. The extended-reality projectile-firing gaming system of claim 1 wherein a player is selectable

through the computer-aided interface and the extended-reality gaming application facilitates communication with the player.

19. The extended-reality projectile-firing gaming system of claim 1 wherein at least one of the virtual objects is sharable through the computer-aided interface with a player.

20. An extended-reality projectile-firing gaming system comprising: a projectile-firing device comprising a projectile repository, a battlefield object, a network configured to provide communication and control connectivity in accordance with at least one protocol to the projectile-firing device and the battlefield object, an extended-reality gaming application configured to control the projectile-firing device and the battlefield object in accordance with a gaming session having a gameplay configuration for providing an extended-reality environment, receive projectile-firing device data from the projectile-firing device, the projectile-firing device data indicating that a projectile was fired, receive battlefield object data from the battlefield object, the battlefield object data indicating that an impact occurred on the battlefield object, and a computer-aided interface for interaction with gameplay control options associated with the gameplay configuration and continually updated in real or near real-time during the gaming session, the computer-aided interface configured to receive commands or instructions associated with the real and virtual objects and corresponding to the gameplay configuration, wherein the battlefield object comprises a smart target comprising a target location, a piezoelectric sensor configured to detect an impact of an object on or within a predetermined distance from the target location and generate a piezoelectric sensor signal corresponding to the detected impact, an infrared sensor configured to detect infrared radiation at a predetermined wavelength and generate an infrared sensor signal, and control circuitry configured to confirm a successful projectile impact on the target location when the object is determined to be a projectile by comparing the piezoelectric sensor signal with a piezoelectric impact signal model trained to detect impact of the projectile and when an impact time of the object and a detection time of the infrared radiation each occurred within a predetermined period of time, generate target location state data indicating that the target location was successfully impacted by the projectile, and transmit the target location state data via the network in accordance with at least one protocol, wherein the extended-reality gaming application is configured to modify the real and virtual objects in real or near real-time in response to the target location state data and display the modified real and virtual objects using the extended-reality device.

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